Chapter 3
Modelling the Power System

To study the stability and to observe how the power system behaves, during normal and abnormal conditions, availability of a simulation model of the considering Power System is very important. To ascertain that the simulation model's performance is identical or approximately equal to the corresponding real time power system's behavior, it has been decided to consider the power system of Sri Lanka of which required data could be collected from the Ceylon Electricity Board (CEB) on request. Therefore a simulation model of the Power System of Sri Lanka (Transmission network—132 and 220 kV) was designed using the software PSCAD/EMTDC.

PSCAD is a power-system simulator for the design and verification of power quality studies, power electronic design, distributed generation, and transmission planning. It was developed by the Manitoba HVDC Research Center and has been in use since 1975. PSCAD is a graphical front end to EMTDC for creating models and analyzing results. In PSCAD, one combines blocks to form a power network. These blocks are actually FORTRAN code, which call for an EMTDC code library to combine them into executable files. Running these files runs the simulations, and the results can be picked up by PSCAD on the run [36–39].

PSCAD is suitable for modelling the power system of Sri Lanka for several reasons. PSCAD has fully developed models of various devices used in the PS of Sri Lanka.

- The library includes models of synchronous machines, turbines, governors, transformers, relays, breakers, cables, and transmission lines. Saturation, magnetizing, and leakage inductances can be disabled or enabled in the rotating machine models. PSCAD also offers tools to simulate various faults on the power system. By providing actual parameters/data where necessary, it is possible to develop models of different power system components.
- The systems of the PSCAD library may consist of electrical and control-type components, which may be interconnected to allow for an all-inclusive simulation study. The control systems modeling function section of the PSCAD library provides a complete set of basic linear and nonlinear control

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components. These components can be combined into larger, more elaborate systems. Outputs from control components can be used to control voltage and current sources, switching signals etc. Control components can also be used for signal analysis, and outputs may be directed to online plots or meters.
- Multiple modules can be built inside a single project, and each module can contain other modules. This provides a hierarchical modeling capability.
- PSCAD has complete models of most of the devices used in the 'simulation model of the PS of Sri Lanka'. Graphing, plotting, and exporting results are easy.

Therefore, PSCAD is suitable for this simulation study.

3.1 Power System Components

For large-scale and complex systems, the mathematical description is nonlinear and high dimensional, consisting of a large number of nonlinear equations. Therefore the analysis and computation of such a system should be started from simple local devices and be completed finally for the complex overall system. Therefore, in modeling of large-scale and complex power systems, these systems are first decomposed into independent basic components, such as:

- Synchronous Generators
- Transformers
- Transmission lines
- Under-ground cables
- Governor
- Turbines etc.

Since these components are already modeled as blocks in PSCAD, it is possible to configure them referring to actual/real time parameters of the power system. Further in configuring these mathematical models corresponding to different components, a sound knowledge of their performance and their effects on the power system must be thoroughly studied. These Models of those components are building-bricks to construct the mathematical model of the whole power systems. Even though the size and structural components of a power system vary from one to another, they all have the same basic characteristics.

3.2 Configuring Power System Components/Mathematical Modeling

For the study of some specific problems, model parameters could be time-variable and variables may not be continuous. To meet the requirement of different computing accuracy, different models could be used. A mathematical model for

qualitative analysis could be simpler than that for quantitative analysis. Since computing accuracy and speed are always two conflicting factors, it is very important to consider whether a very high accuracy is required with the results, when a power system model is established [40].

There are two major issues in mathematical modelling [40]. This can be brief according to the Fig. 3.1.

As demonstrated in Fig. 3.1, Issues of mathematical modelling of a power system/power system components can be identified as:

- To describe a subject under investigation mathematically in the form of equations
  - Analytical method

    to derive those mathematical descriptions by using special knowledge and theory about the subject
  - Experimental method

    To identify those mathematical descriptions by carrying out experiments or using data obtained from its operation—the method of system identification in control theory
- To obtain parameters of the mathematical description of the subject:
  - Based on design parameters (theoretical derivation)

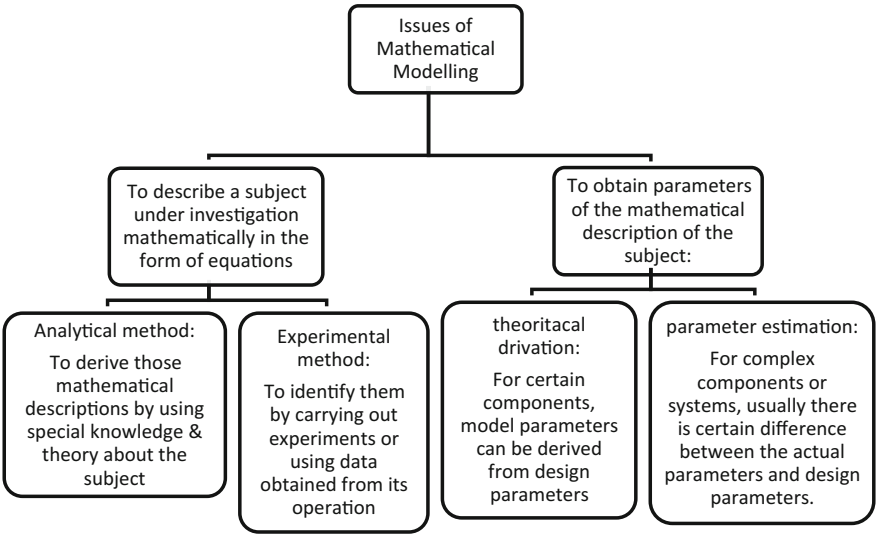


Fig. 3.1 Issues of mathematical modelling of a power system/power system components

For simple components, model parameters can be derived from design parameters according to certain physical (e.g. mechanical or electrical) principles; e.g. transmission line modeling

– Based on Parameter estimation

For complex components or systems, usually there is certain difference between the actual parameters and design parameters; e.g. generator modelling (these generator parameters can be affected by variations of the power system operating conditions, saturation and a series of complex conversion processes among mechanical, electrical, magnetic and thermal energy). This is known as 'parameter estimation' which is one of the methods in system identification.

3.2.1 Transmission Lines

Transmission lines are the main corridor of power transmission in a power system. The '*Generation and Transmission network*' as in 2011 in the power system of Sri Lanka [19] has been referred, **Appendix-D**.

Conductor Types Available in the Power System
It has been identified that different types of conductors are utilized in transmission of power [19]. All are 3-phase circuits. Some of them are single circuits while the others are double circuits. Table 3.1 gives a detailed list of conductors used in the transmission network and their corresponding per unit values of +ve sequence resistance, +ve sequence reactance and +ve sequence susceptance. These values were calculated based on the existing transmission line data given in [41].

Sample Calculation
Calculation of a p.u. value corresponding to a 220 kV transmission line:
Calculation of Z_{base} corresponding to 220 kV transmission line:

$$Z_{base} = \frac{V_{base}^2}{MVA_{base}}$$

where,

$$MVA_{base} = 100 \text{ and } V_{base} = 220\,\text{kV}$$

$$Z_{base} = \frac{220^2}{100} = 484\,\Omega$$

$$\frac{Resistance}{circuit} in\ p.u. = \frac{resistance\ of\ the\ circuit}{Z_{base}}\Omega$$

Table 3.1 Conductor types used in the transmission network of Sri Lanka

Conductor	Voltage capacity (kV)	+ve sequence resistance pu/m (Ω)	+ve sequence reactance pu/m (Ω)	+ve sequence susceptance pu/m (Ω)
lynx	132	1.016×10^{-6}	2.353×10^{-6}	4.867×10^{-7}
Oriole	132	1.0962×10^{-6}	2.5022×10^{-6}	4.8398×10^{-7}
Zebra	132	4.332×10^{-7}	2.202×10^{-6}	5.193×10^{-7}
Goat	132	5.739×10^{-7}	2.2383×10^{-6}	5.1244×10^{-7}
Tiger	132	1.4176×10^{-6}	2.4047×10^{-6}	4.7477×10^{-7}
2 × Zebra	132	2.1667×10^{-7}	1.7259×10^{-6}	6.524×10^{-7}
2 × Goat	132	2.864×10^{-7}	1.7389×10^{-6}	6.5256×10^{-7}
Zebra	220	1.5597×10^{-7}	8.5371×10^{-7}	1.3353×10^{-6}
2 × Zebra	220	7.8×10^{-8}	6.2133×10^{-7}	1.8124×10^{-6}
2 × Goat	220	1.0312×10^{-7}	6.276×10^{-7}	1.7937×10^{-7}

The transmission line from Randenigala to Rantembe has been taken as a sample for model calculations [19].

Type of circuit	Single circuit
Names of transmission lines	Rand-Rant
Steady state frequency	50.0 Hz
Length of transmission line	3.1 km
Number of conductors (per circuit)	3 nos.
Conductor type	2 × Zebra
Voltage capacity	220 kV

Referring to the data given in [41], +ve sequence resistance (R), reactance (X) and suceptance (Y) values of transmission lines were calculated. Table 3.2 gives the corresponding data used for the transmission line 'Rand-Rant' parameter calculation.

Calculation of +ve sequence resistance, +ve sequence reactance and +ve sequence susceptance:

$$+ve\ sequence\ resistance = \frac{0.00024}{3.1 \times 10^3}\ \text{pu/m} \cong 7.8 \times 10^{-8}\ \text{pu/m}$$

Table 3.2 RXY values as given in [41], for the transmission line 'Rand-Rant'

Conductor	Voltage capacity (kV)	Resistance (R)/circuit in p.u. (Ω)	Reactance (X)/circuit in p.u. (Ω)	Susceptance (Y)/circuit in p.u. (Ω)
2 × Zebra	220	0.00024	0.00196	0.00605

$$+ve\ sequence\ reactance = \frac{0.00196}{3.1 \times 10^3}\ \text{pu/km} \cong 6.2133 \times 10^{-7}\ \text{pu/m}$$

$$+ve\ sequence\ susceptance = \frac{0.00605}{3.1 \times 10^3}\ \text{pu/km} \cong 1.8124 \times 10^{-6}\ \text{pu/m}$$

Values Used with PSCAD Window
In configuring the transmission lines, the Bergeron model was used since,

- It is useful for studies to get the correct steady state impedance/admittance of the transmission line at a specified frequency, and it is not used in analysis of transient studies or harmonic behavior.
- It is a very simple and constant frequency model based on travelling waves [37].

Figure 3.2 shows the PSCAD windows where the corresponding transmission line parameters were entered in configuring the transmission line—'Rand-Rant'. Figure 3.3 demonstrates as it appears in the above transmission line has been simulated in the Sri Lanka power system simulation model.

Figure 3.3 is a small part of the large transmission network of the power system of Sri Lanka that was extracted from the simulation model done. It shows how the bus-bars of power stations randenigala and Rantembe are linked through the transmission line 'Rand-Rant.'

3.2.2 Under-Ground Cables

In certain places of the power system, it has been identified the power transmission is done through underground cables. These cables are with different current capacities [42, 43]. In configuring these under-ground cables in the PSCAD software, factors such as

- their construction criteria
- insulation materials used
- how they are laid down etc. are considered.

Cable Types Available in the Power System
With reference to [19], the types of cables used for power transmission in the power system of Sri Lanka are:

- XLPE 150
- Cu 500
- 1000 XLPE
- 800 XLPE
- Cu 350

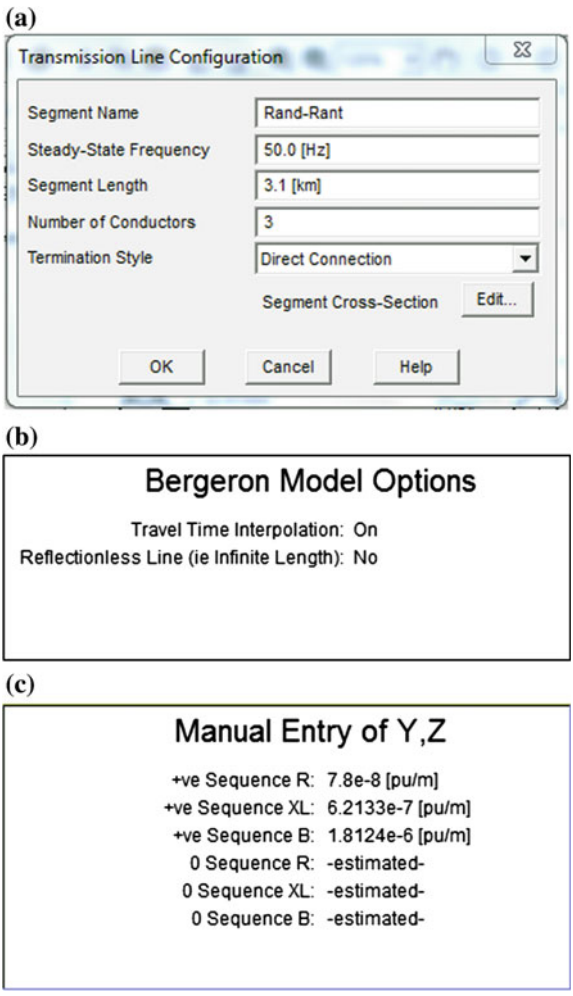


Fig. 3.2 PSCAD windows corresponding to transmission line from Randenigala to Rantembe named as 'Rand-Rant' with it's **a** line model general data; **b** Bergeron model options; **c** manual entry of Y, Z

Unlike over-head transmission conductors, the under-ground cables are comprised with several layers of insulation and protective materials. With reference to [42], XLPE cable components are demonstrated in Fig. 3.4.

It has been assumed that the conducting materials in all cables are copper. Details of the cable physical parameters, resistivity of metals used and permittivity of insulating materials used are presented in Tables 3.3, 3.4, and 3.5 respectively [42, 43].

The voltage of a cable circuit is designed in accordance with the following principles: Eg: U_0/U (U_m): 130/225 (245)

where,

U_0 = 130 kV, phase to ground voltage

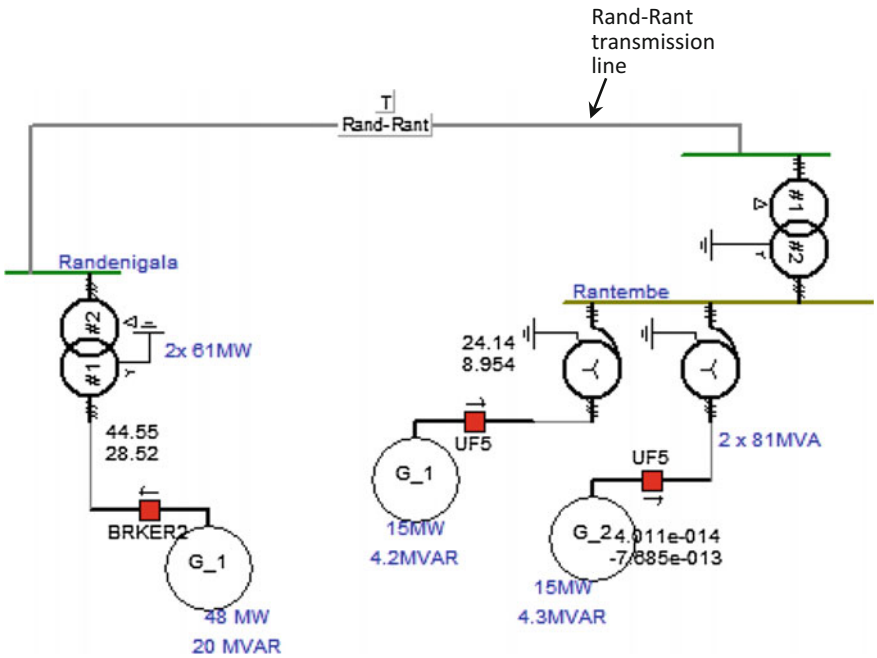


Fig. 3.3 A part of the power system comprising the 'Rand-Rant' transmission line that links Randenigala and Rantembe power-stations

U = 225 kV, rated phase to phase voltage
Um = 245 kV, highest permissible voltage of the grid

Values Used with PSCAD Window

Figure 3.5 is a part of the transmission network of the simulation model of the power system of Sri Lanka. The cable that links Colombo Fort substation (Sub-F) and the Kelanitissa power station bus-bars are shown in it. Even though the Fig. 3.4 demonstrates several layers of insulations and screen/sheath in addition to the conductor, for this simulation model it has been assumed that,

- Each cable is comprised of conductor, insulator-1, insulator-2 and sheath only.
- All three cables are placed horizontally on the same level of the ground
- Span between two cables (center to center) = 2 × cable diameter [42].
- Each cable carries a current of single phase (for three phases, 3 nos. separate cables have been used)

Accordingly, how such a cable (Keltissa-Col_F) has been configured in PSCD is demonstrated in Fig. 3.6.

Cable configuration window, Line model general data and Bergeron Model Options corresponding to 'Keltissa-Col_F' are demonstrated in Fig. 3.6a, b and c respectively. Figure 3.7 shows how the above 132 kV under-ground cable have been laid down in earth.

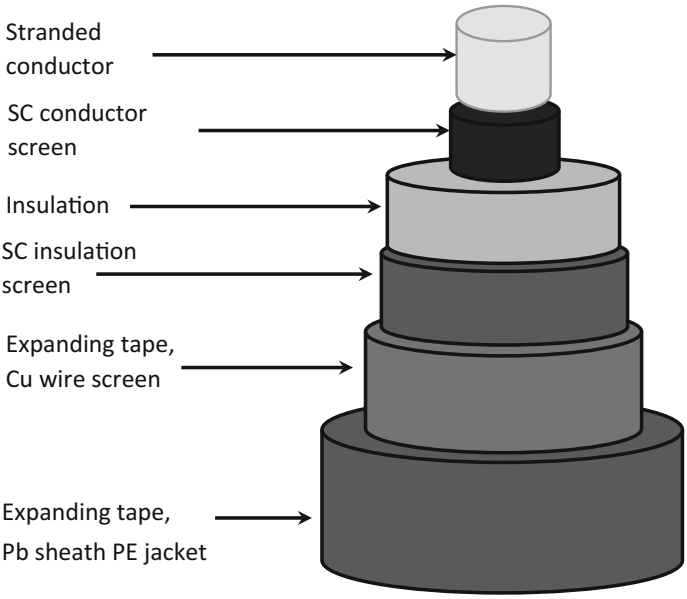


Fig. 3.4 Different layers of insulation and protective materials of an underground cable [42]. Where, SC—semiconductor, Cu—copper, Pb—lead, PE—poly ethylene

Table 3.3 Physical parameter data used for configuring the cables in the simulation model

Name of the cable	1000 R-Cu 130/225 (245) kV	800 R-Cu 130/225 (245) kV	400 R-Cu 130/225 (245) kV	500 R-Cu 76/132 (145) kV	400 R-Cu 76/132 (145) kV
Conductor diameter (mm)	38.2	34.7	23.3	26.7	23.2
Thickness of insulation (mm)	14.8	14.8	21.6	16.2	17.1
Al screen: out-side diameter of cable (mm)	85	82	85	77	75
Cu wire/Al sheath: out-side diameter of cable (mm)	88	84	87	79	77

Table 3.4 Resistivity of metals used in the cables

Material	Resistivity at 20 °C (mΩ)
Aluminum (Al)	2.803×10^{-8}
Copper (Cu)	1.724×10^{-8}
Lead (Pb)	21.4×10^{-8}

Table 3.5 Permittivity of insulation materials used in the cables

Material	Permittivity at 50 Hz
XLPE	2.35
Impregnated paper (high density)	3.8
Impregnated paper (low density)	3.3

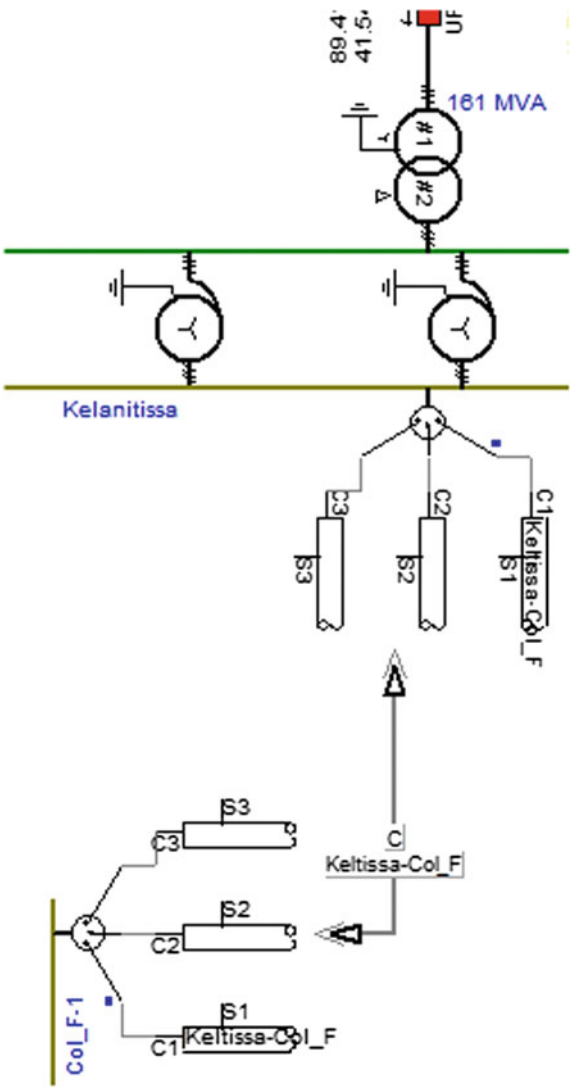


Fig. 3.5 The cable that links Colombo Fort substation and Kelanitissa power station bus-bars, which appears in the transmission network of the simulation model of the power system of Sri Lanka

The resistivity of soil varies widely at different locations as the type of soil changes. At some locations, the soil can be very nonhomogeneous with multilayer soil structure, not far away from the earth surface. Often, there are several layers of

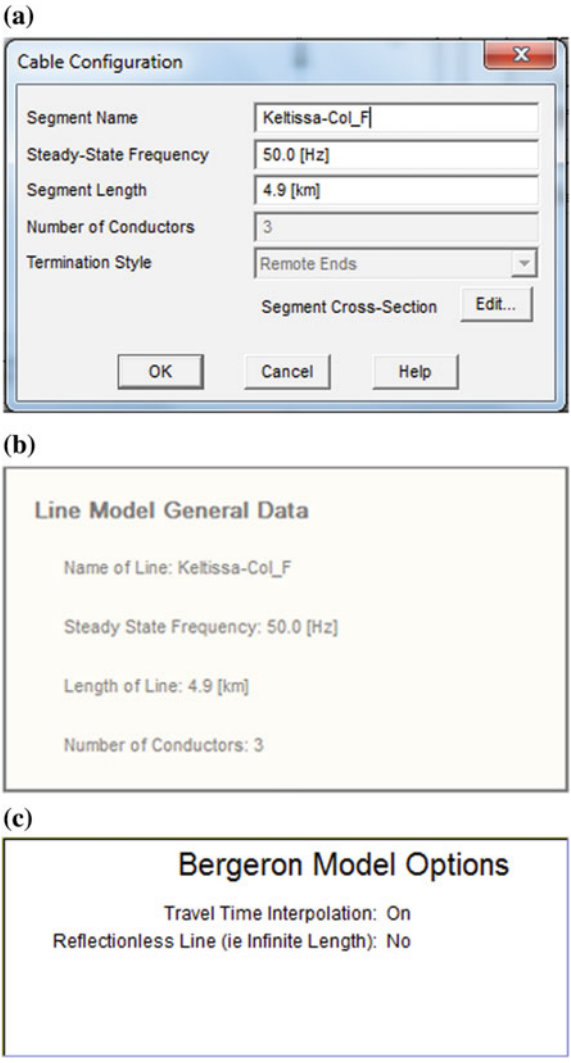


Fig. 3.6 PSCAD windows corresponding to power transmission cable from Kelanitissa to Colombo Fort substation named as 'Keltissa-Col_F' with it's **a** cable configuration window; **b** line model general data; **c** Bergeron model options

soil made up of loam, sand, clay, gravel, and rocks. The layers may be roughly horizontal to the surface or inclined at an angle to the surface. The resistivity also fluctuates seasonally due to changes in rainfall and temperature. The impact of temperature is only important near and below freezing point [44]. Hence it has been taken the (average) ground resistivity as 100 Ωm at a depth of 1.3 m from ground level, for the entire simulation.

Since there are several such cables cater for power transmission in the power system net-work, corresponding data which was used for cable configuration is available in Table 3.6.

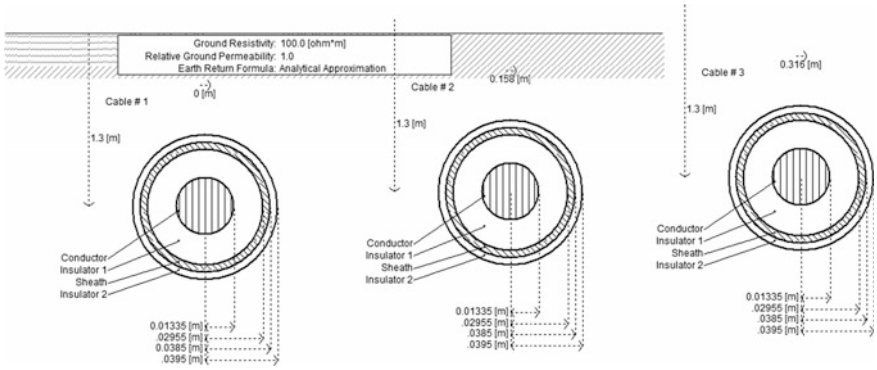


Fig. 3.7 132 kV under-ground cable lay-out corresponding to 'Keltissa-Col_F'

Table 3.6 Data used for configuring under-ground/submarine cables

Name of the cable	Depth of the cable from ground level (m)	Span between two cables (m)	Conductor diameter (m)	Thickness of insulator-1 (XLPE) (m)	Thickness of sheath (m)	Thickness of insulator-2 (m)
XLPE 150	1.3	0.174	0.0233	0.0216	9.25×10^{-3}	1×10^{-3}
Cu 350	1.3	0.154	0.0232	0.0171	8.8×10^{-3}	1×10^{-3}
Cu 500	1.3	0.158	0.0267	0.0162	8.825×10^{-3}	1×10^{-3}
XLPE 800	1.3	0.168	0.0347	0.0148	8.85×10^{-3}	1×10^{-3}
XLPE 1000	1.3	0.176	0.0382	0.0148	10^{-3}	1×10^{-3}

3.2.3 Transformers

Transformers are the devices that permit high voltage transmission and this drastically reduces losses. At corresponding grid substations, transmission voltage/s is step down to a distribution voltage level. Power transportation is accomplished through distribution systems. This includes small networks of radial or ring-main configuration and voltages are stepped down to end-user levels once again through transformers [14].

Transformer Types Available in the Power System

In this simulation generator transformers and transformers at grid sub-stations which step down 220–132 kV were considered. The generator transformers are with different capacities. The primary voltage is the voltage at which the electricity is generated. Some of the grid sub-stations are comprised with auto transformers that step down the voltage to 132 and 33 kV simultaneously. Since the intension of

the simulation is to model the transmission network, the 33 kV, which is at primary distribution level, was not considered.

Sample Calculation

The calculations were carried out referring to [45], pp. 36–37. Typical per-unit (pu) values of transformers are given in Tables 3.7 and 3.8. For the transformer capacity ranges given in Table 3.7, the leakage reactance (pu) on primary and secondary windings can be considered as equal for design purposes. In the same way, the copper loss on each side has been considered as equal.

Example:

Sample calculation of generator transformer parameters of Kotmale power station (less than 100 MVA):

Transformer rating = 90 MVA
Primary winding—star
Secondary winding—delta
Primary voltage—13.8 kV
Secondary voltage—220 kV

$$+\text{ve sequence leakage reactance of primary}, x_1 = \left(\frac{+0.03}{100} \times 90\right) + 0.03$$
$$= 0.057\,\text{pu}$$

$$+\text{ve sequence leakage reactance of secondary referring to primary}, x_1' = \left(\frac{13.8}{220}\right)^2 \times 0.057\,\text{pu}$$
$$= 0.00393 \times 0.057\,\text{pu}$$
$$= 0.0002\,\text{pu}$$

$$\therefore +\text{ve sequence leakage reactance} = 0.057 + 0.0002\,\text{pu}$$
$$= 0.057\,\text{pu}$$

Table 3.7 Typical per-unit values of transformers [45]

Circuit element	Typical per-unit values	
	3–250 kVA	1–100 MVA
R_1 or R_2	0.009–0.005	0.005–0.002
X_{f1} or X_{f2}	0.008–0.025	0.03–0.06
X_m	20–30	50–200
R_m	20–50	100–500
I_0	0.05–0.03	0.02–0.005

Table 3.8 Per-unit values of transformer parameters [45]

S_n	kVA	1	10	100	1000	400,000
E_{np}	V	2400	2400	12470	69,000	13,800
E_{ns}	V	46	347	00	6900	424,000
I_{np}	A	0.417	4.17	8.02	14.5	29,000
I_{ns}	A	2.17	28.8	17	145	943
Z_{np}	Ω	5760	576	1555	4761	0.4761
Z_{ns}	Ω	211.6	12.0	3.60	47.61	449.4
R_1(pu)	–	0.0101	0.0090	0.0075	0.0057	0.00071
R_2(pu)	–	0.0090	0.0079	0.007	0.0053	0.00079
X_{f1}(pu)	–	0.0056	0.0075	0.0251	0.0317	0.0588
X_{f2}(pu)	–	0.0055	0.0075	0.0250	0.0315	0.06601
X_m(pu)	–	34.7	50.3	96.5	106	966
R_m(pu)	–	69.4	88.5	141.5	90.7	66
I_0(pu)	–	0.032	0.023	0.013	0.015	0.0018

$$\text{Copper loss resistance of primary, } r_1 = \left(\frac{-0.003}{100} \times 90\right) + 0.005 = 0.0027\,\text{pu}$$

$$\text{Copper loss resistance of secondary referring to primary, } r_1' = \left(\frac{13.8}{220}\right)^2 \times 0.0027\,\text{pu} = 0.00393 \times 0.0027\,\text{pu} = 0.000010\,\text{pu}$$

$$\therefore \text{total copper loss resistance, } = 0.0027 + 0.00001\,\text{pu} = 0.00271\ \text{pu}$$

$$\text{Air core reactance, } = 2 \times (+ve\ sequence\ reactance) = 2 \times 0.057\,\text{pu} = 0.114\,\text{pu}$$

The data given in Table 3.8 corresponds to transformer ratings from 1 kVA to 400 MVA. So calculation of transformer parameters whose ratings are above 100 MVA was carried out based on the data of the Table 3.8.

Sample calculation of parameters, of a transformer that has a rating above 100 MVA:

This is located at Biyagama-grid substation.
Transformer rating = 250 MVA
Transformer type—auto-transformer

Primary winding—star
Secondary winding—delta
Primary voltage—220 kV
Secondary voltage—132 kV

$$+\text{ve sequence leakage reactance of primary}, x_1 = \left(\frac{+0.0588 - 0.0317}{400}\right) \times 250 + 0.0317 = 0.04864\,\text{pu}$$

$$+\text{ve sequence leakage reactance of secondary referring to primary}, x_2' = \left(\frac{220}{132}\right)^2 \times 0.04864\,\text{pu} = 0.1351\,\text{pu}$$

$$\therefore \text{equivalent} + \text{ve sequence leakage reactance}, x_{eq} = x_1 + x_2'\,\text{pu} = 0.0486 + 0.1351\,\text{pu} = 0.18375\,\text{pu}$$

$$\text{Copper loss resistance of primary}, r_1 = \left(\frac{0.0007 - 0.0057}{400}\right) \times 250 + 0.0057 = 0.00258\,\text{pu}$$

$$\text{Copper loss resistance of secondary referring to primary}, r_2' = \left(\frac{220}{132}\right)^2 \times 0.00258\,\text{pu} = 0.00717\,\text{pu}$$

$$\therefore \text{total copper loss resistance}, = r_1 + r_2'\,\text{pu} = 0.00258 + 0.00717\,\text{pu} = 0.00975\,\text{pu}$$

$$\text{Air core reactance}, = 2 \times (+\textit{ve sequence reactance}) = 2 \times 0.057\,\text{pu} = 0.114\,\text{pu}$$

Values Used with PSCAD Window

There are three transformers each with a capacity of 90 MVA, installed in the Kotmale power station. Figure 3.8 shows them as they appear in the PSCAD simulation.

PSCAD windows corresponding to a transformer configuration and saturation for the generator transformers at Kotmale power plant are shown in Fig. 3.9a, b.

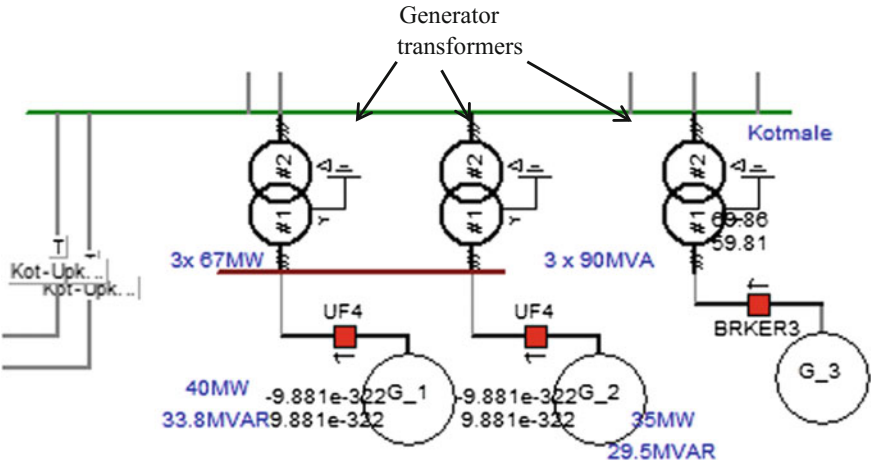


Fig. 3.8 Generator transformers located at the Kotmale power station, as they appear in the PSCAD simulation model

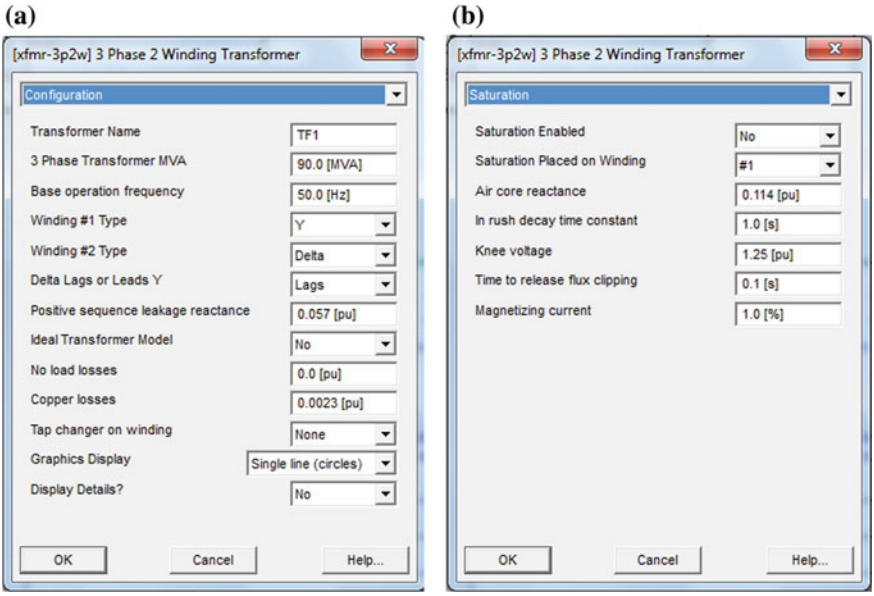


Fig. 3.9 PSCAD generator transformer windows of **a** configuration; **b** saturation; corresponding to the Kotmale power station

3.2.4 Generators

An electric generator is a device designed to take advantage of electromagnetic induction in order to convert movement into electricity. It is designed to obtain an induced current in a conductor (or set of conductors) as a result of mechanical movement, which is utilized to continually change a magnetic field near the conductor. The generator thus achieves a conversion of one physical form of energy into another—energy of motion into electrical energy—mediated by the magnetic field that exerts forces on the electric charges [46].

With reference to [6], synchronous generators can be roughly classified as shown in Fig. 3.10.

Further, a generating unit can be demonstrated as shown in the Fig. 3.11.

- Electricity is produced by a synchronous generator driven by a prime mover, usually a turbine or a diesel engine. The turbine is equipped with a turbine governor to control either the speed or the output power according to a preset power–frequency characteristic.
- Through a transmission network, the generated power is transmitted over to load-centres after stepping up the generated voltage (by a generator transformer).
- The DC excitation (or field) current, required to produce the magnetic field inside the generator, is provided by the exciter. The excitation current, and consequently the generator's terminal-voltage, is controlled by an automatic voltage regulator (AVR).

In modelling a real time generator, it is very important to have a clear idea about how the energy in mechanical form is converted to electrical form, in order to

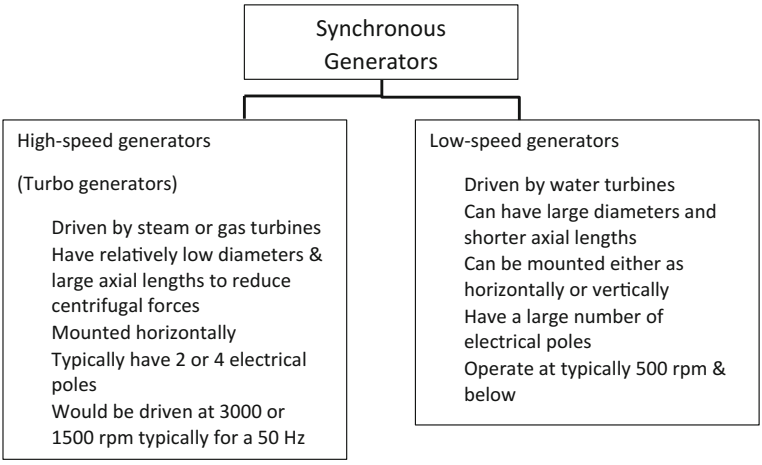


Fig. 3.10 Classification of synchronous generators referring to their speeds

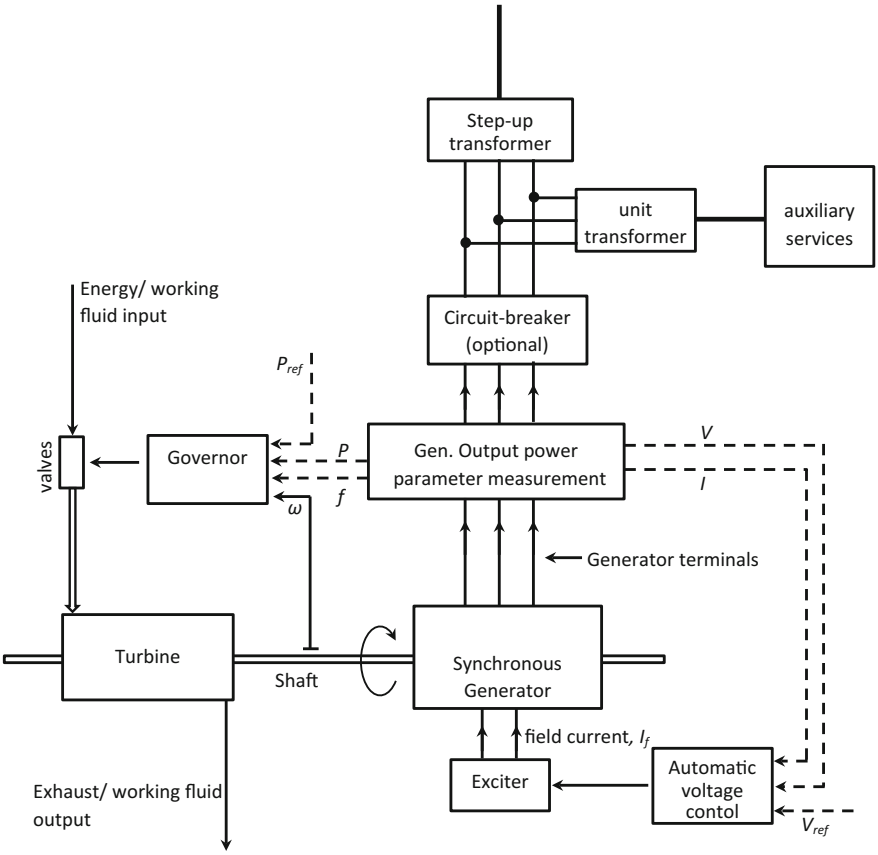


Fig. 3.11 Configuration of a generating unit [6]

maintain some specific frequency, while the electricity in demand varies. This would be very helpful in configuring the machine so that it behaves like a specific real-time generator. With reference to [11] and [13], basic generator model was developed.

Defining electrical-mechanical system terms:

ω	rotational speed (rad/s)
α	rotational acceleration
δ	phase angle of a rotating machine
T_{net}	net acceleration torque in a machine
T_{mech}	mechanical torque exerted on the machine by the turbine
T_{elec}	electrical torque exerted on the machine by the generator
P_{net}	net accelerating power
P_{mec}	mechanical power input
P_{elec}	electrical power output

I moment of inertia for the machine
M angular momentum of the machine

- All these quantities (except phase angle) will be in per unit on the machine base, or in the case of ω, on the standard system frequency base.
- It is considered deviations of quantities about the steady state values.
- All steady-state or nominal values may be denoted with a sub-script '0' (e.g., ω_0, T_{net_0}).
- All deviations from nominal may be denoted by a 'Δ' (e.g., $\Delta\omega, \Delta T_{net}$).

$$\begin{aligned} I\alpha &= T_{net} \\ M &= \omega I \\ P_{net} &= \omega T_{net} = \omega(I\alpha) = M\alpha \end{aligned} \tag{3.1}$$

Consider a rotating machine with a steady speed of ω_0 and phase angle δ_0. Due to various electrical or mechanical disturbances, the machine will be subjected to differences in mechanical and electrical torque, causing it to accelerate or decelerate.

Consider a deviation of speed $\Delta\omega$ and a deviation in phase angle $\Delta\delta$, from nominal.

$$\omega = \omega_0 + \alpha t$$

$$\Delta\delta = \int (\omega_0 + \alpha t)dt - \int (\omega_0)dt$$

Machine absolute phase angle

Phase angle of reference axis

$$= \omega_0 t + \frac{1}{2}\alpha t^2 - \omega_0 t$$

$$= \frac{1}{2}\alpha t^2$$

Deviation from nominal speed,

$$\begin{aligned} \Delta\omega = \alpha t &= \frac{d}{dt}(\Delta\delta) \\ \therefore T_{net} = I\alpha &= I\frac{d}{dt}(\Delta\omega) = \frac{d^2}{dt^2}(\Delta\delta) \end{aligned} \tag{3.2}$$

With reference to Appendix-C the relationship between net acceleration power and the electrical and mechanical power is:

$$P_{net} = P_{mech} - P_{elec}$$

This can be written as a sum of the steady-state value and the deviation term,

$$P_{net} = P_{net_0} - \Delta P_{net}$$

where

$$P_{net_0} = P_{mech_0} - P_{elec_0}$$
$$\Delta P_{net} = \Delta P_{mech} - \Delta P_{elec}$$

Then,

$$P_{net} = (P_{mech_0} - P_{elec_0}) + (\Delta P_{mech} - \Delta P_{elec}) \tag{3.3}$$

Similarly for torques,

$$T_{net} = (T_{mech_0} - T_{elec_0}) + (\Delta T_{mech} - \Delta T_{elec}) \tag{3.4}$$

Since Eq. (3.1) $\Rightarrow$,

$$P_{net} = \omega T_{net}$$

$$P_{net} = P_{net_0} + \Delta P_{net} = (\omega_0 + \Delta\omega)(T_{net_0} + \Delta T_{net})$$

From Eqs. (3.3) and (3.4)

$$(P_{mech_0} - P_{elec_0}) + (\Delta P_{mech} - \Delta P_{elec}) = (\omega_0 + \Delta\omega)[(T_{mech_0} - T_{elec_0}) + (\Delta T_{mech} - \Delta T_{elec})]$$

Assume that the steady-state quantities can be factored out since

$$P_{mech_0} = P_{elec_0}$$

and

$$T_{mech_0} = T_{elec_0}$$

and further assume that the second-order terms involving products of $\Delta\omega$ with ΔT_{mech} and ΔT_{elec} can be neglected. Then,

$$(\Delta P_{mech} - \Delta P_{elec}) = \omega_0(\Delta T_{mech} - \Delta T_{elec}) \tag{3.5}$$

As shown in Eq. (3.2), the net torque is related to the speed change as follows:

$$(T_{mech_0} - T_{elec_0}) + (\Delta T_{mech} - \Delta T_{elec}) = I\frac{d}{dt}(\Delta\omega) \tag{3.6}$$

Then, since $T_{mech0} = T_{elec0}$, by combining Eqs. (3.5) and (3.6),

$$\begin{aligned}(\Delta P_{mech} - \Delta P_{elec}) &= \omega_0 I\frac{d}{dt}(\Delta\omega)\\ &= M\frac{d}{dt}(\Delta\omega)\end{aligned}$$

$$\mathcal{L}(\Delta P_{mech} - \Delta P_{elec}) = \mathcal{L}\left(M\frac{d}{dt}(\Delta\omega)\right)$$

∴ In Laplace transform operator notation,

$$\Delta P_{mech} - \Delta P_{elec} = Ms(\Delta\omega)$$

This relationship between mechanical and electrical power and speed change can be given in a block-diagram as shown in Fig. 3.12.

The loads on a power system consist of a variety of electrical devices such as,

- purely resistive
- motor loads with variable power–frequency characteristics

Since motor loads are a dominant part of the electrical load, the effect of the change in load due to the change in frequency can be given by:

$$\Delta P_{L(freq)} = D\Delta\omega \; or \; D = \frac{\Delta P_{L(freq)}}{\Delta\omega}$$

∴ The net change in P_{elec} in Fig. 3.12, (explains in Eq. (3.6)) is:

$$\Delta P_{elec} = \underbrace{\Delta P_L}_{Nonfreqency-sensitive\ load\ change} + \underbrace{D\Delta\omega}_{Freqency-sensitive\ load\ change}$$

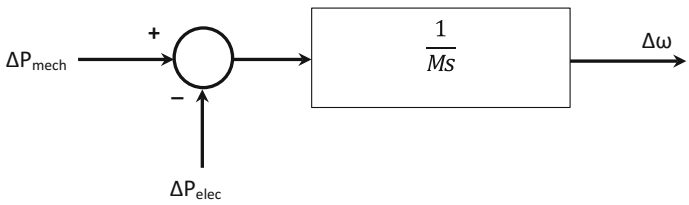


Fig. 3.12 Relationship between mechanical and electrical power and speed change

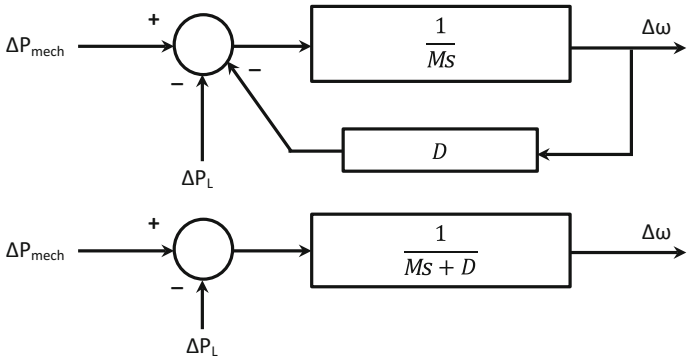


Fig. 3.13 Block diagrams demonstrating the effect of change in frequency sensitive and non-frequency sensitive load

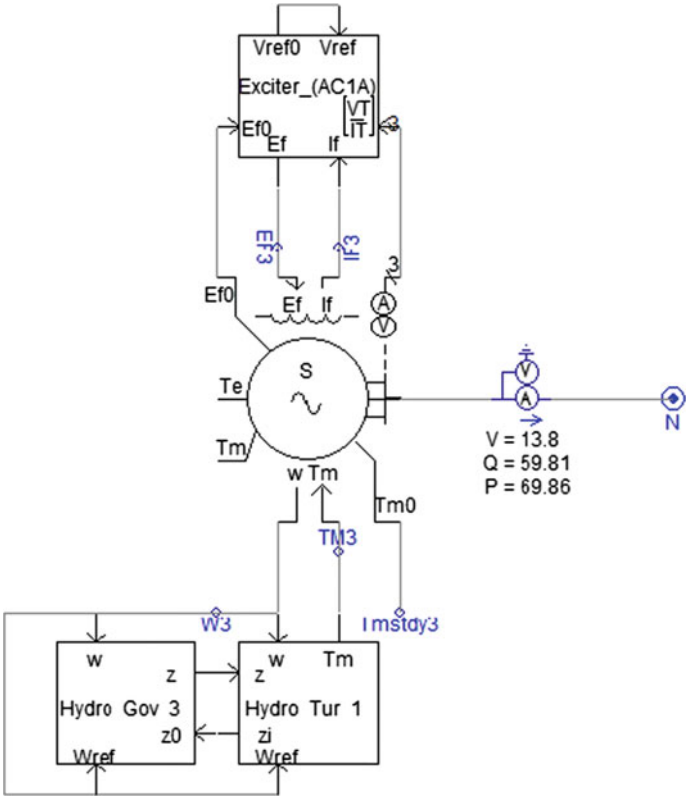


Fig. 3.14 A generator unit at Kotmale power station with its hydro turbine, governor and exciter

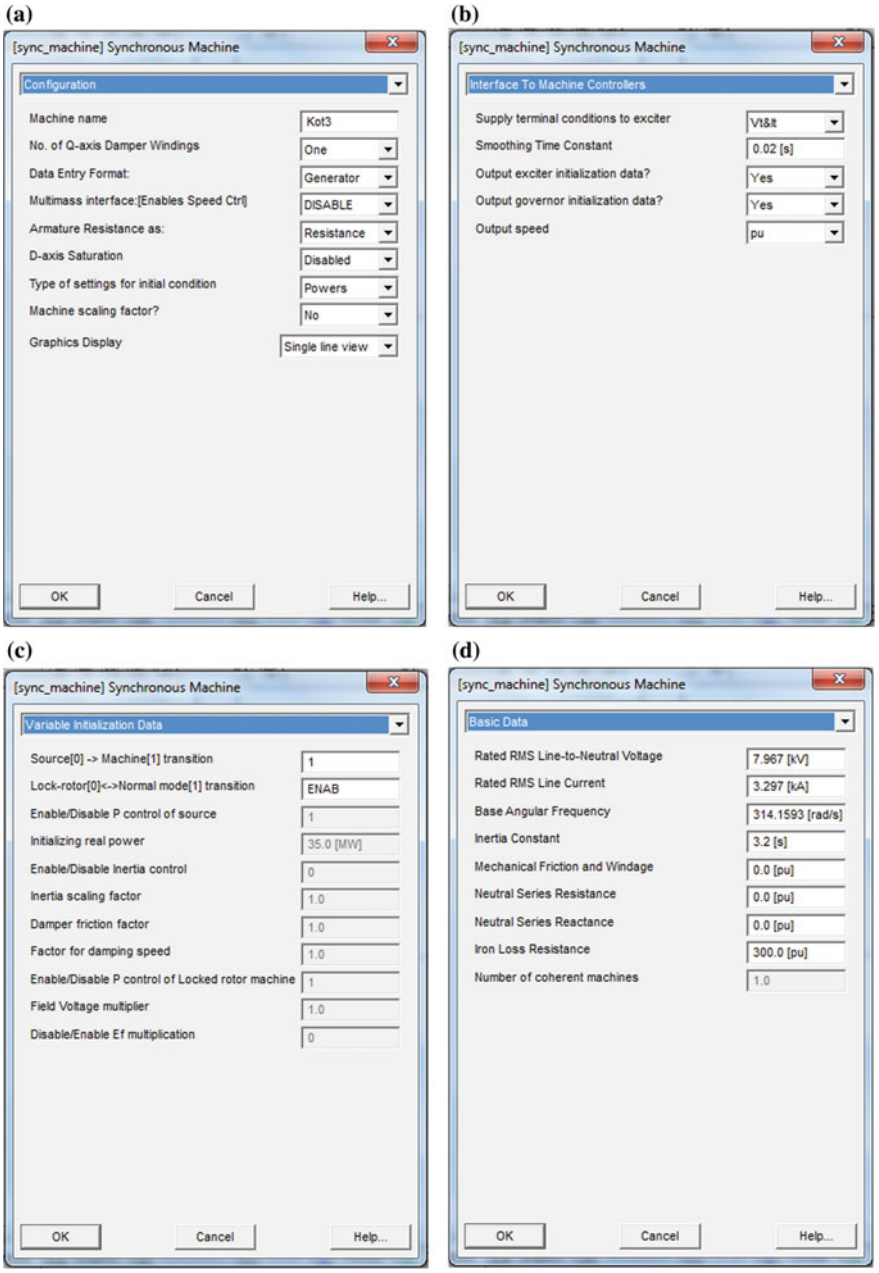


Fig. 3.15 Values used with PSCAD window corresponding to Kotmale generator; **a** synchronous machine configuration; **b** interface to machine controllers; **c** variable initialization data; **d** basic data; **e** initial conditions; **f** initial conditions if starting as a machine; **g** output variable names and **h** power output of the simulated generator 'Kotmale-3'

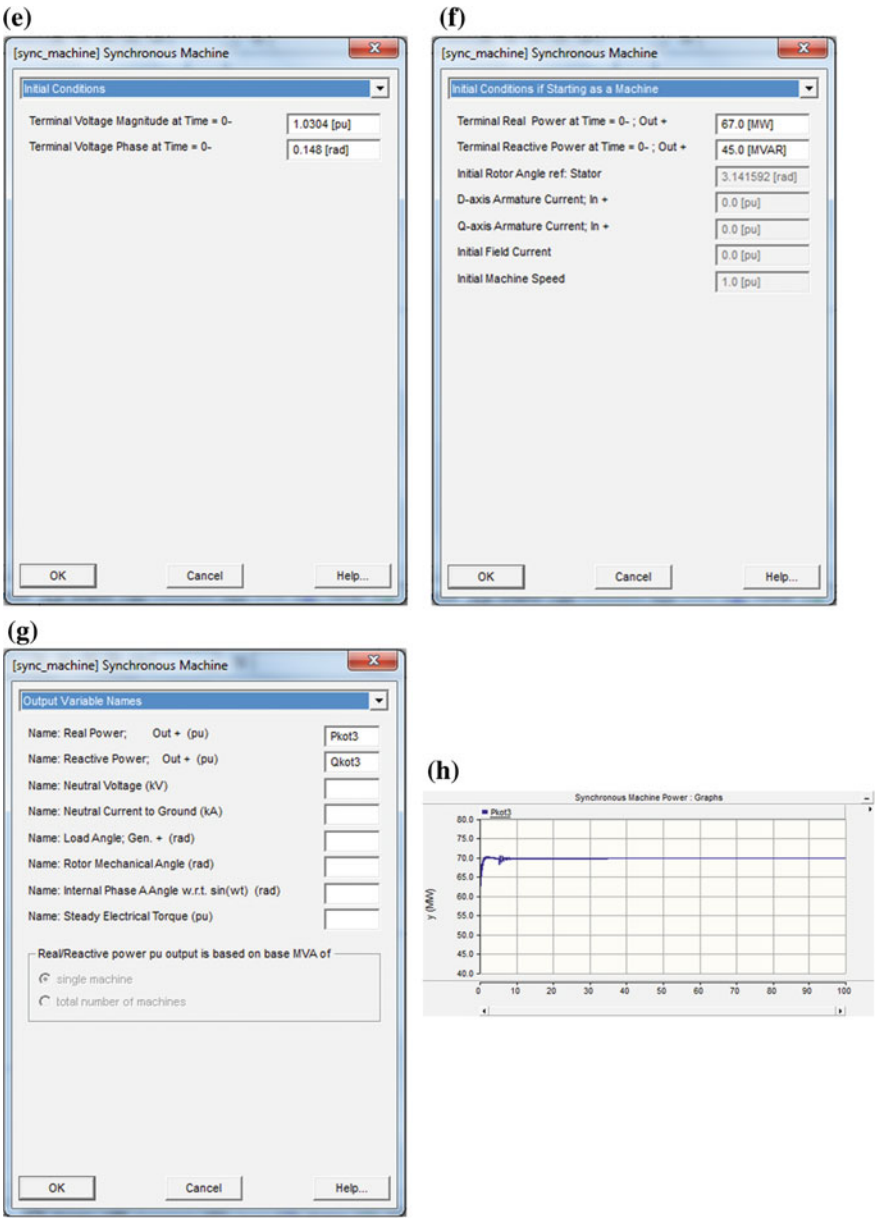


Fig. 3.15 (continued)

Table 3.9 Inertia constants of different types of generators

Generator type	Range of inertia constant (s)
Hydraulic unit (water wheel generator)	2–4
Thermal unit:	
• Condensing (1800 rpm) • Condensing (3000 rpm) • Non-condensing (3000 rpm)	9–6 7–4 4–3

This describes the block-diagram shown in the Fig. 3.13.

Generator Types Used in Modeling the Power System
In this power system simulation model, basically two types of generators were used. They are:

- Hydro-generators
- Steam-turbine generators

All the thermal power generators which were committed at the instant of the corresponding load-flow simulation were represented with steam turbine generators.

Simplified Schematic Diagram and Corresponding Control System
With reference to the PSCAD simulation model, the Fig. 3.14 shows a hydro-generator unit located at Kotmale power station, (Kotmale3) with its corresponding governor, turbine and exciter units.

Values Used with PSCAD Window
Figure 3.15 with (a) Synchronous machine configuration, (b) Interface to machine controllers, (c) Variable initialization data, (d) Basic data, (e) Initial conditions, (f) Initial conditions if starting as a machine, (g) Output variable names and (h) power output of the simulated generator 'Kotmale-3' show the configuration windows and graph of power generated, corresponding to Kotmale hydro-generator.

Sample Calculation
Sample calculation of parameters, of the generator Kotmale3:

Rated (active) power = 67 MW
Rated (reactive) power = 78.8 MVA
Rated output voltage = 13.8 kV

$$\text{Rated RMS line to neutral voltage} = \frac{13.8}{\sqrt{3}} = 7.967\,\text{kV}$$

$$\text{Rated RMS line current} = \frac{78.8 \times 10^6}{\sqrt{3} \times 13.8 \times 10^3} = 3.297\,\text{kA}$$

Table 3.10 Inertia constants of generators obtained from the CEB

Station/generator	Inertia constant (H s)
Barge	1.62
KCCPgas	8
KCCPsteam	4
Sapugaskanda-1	3.2
Sapugaskanda-2	3.2
Sapugaskanda-3	3.2
Kerawalapitiya-1	–
Kerawalapitiya-2	–
Kerawalapitiya-3	–
KHD (Asia power)-1	0.997
KHD (Asia power)-2	0.997
Puttalam Coal	–
Upper Kotmale	–
Heledanavi-1	1.3
Heledanavi-2	1.3
Kotmale-3	3.02
Polpitiya-1	2.84
Polpitiya-2	2.84
Canyon-1	3.8
Canyon-2	3.8
Kukule-1	3
Kukule-2	3
New Laxapana-1	3.3
New Laxapana-2	3.3
Victoria-1	3.45
Victoria-2	3.45
Victoria-3	3.45
Ukuwela	2.72
Randenigala	3.65
Wimalasurendra	3.2
Laxapana-1	3.17
Laxapana-2	2.45
Samanalawewa-1	3.4
Samanalawewa-2	3.4
Bowatenna	4
Rantembe-1	2.62
Embilipitiya	1.1

Table 3.11 Inertia constants used for the units considered in the simulation

Station/generator	Unit capacity (MW)	Generator type	Inertia constant (H s)
Barge	60	Thermal (diesel)	3.14
KCCPgas	161	Thermal (gas)	3.79
KCCPsteam	81	Thermal (steam)	3.14
Sapugaskanda-1	9 MW × 4	Thermal (diesel: coherent—4 units.)	2.53
Sapugaskanda-2	9 MW × 4	Thermal (diesel: coherent—4 units.)	2.53
Sapugaskanda-3	18 MW × 4	Thermal (diesel: coherent—4 units.)	2.65
Kerawalapitiya-1	100 MW	Thermal (gas)	3.62
Kerawalapitiya-2	100 MW	Thermal (gas)	3.62
Kerawalapitiya-3	100 MW	Thermal (gas)	3.62
KHD (Asia power)-1	6.375 MW × 4	Thermal (diesel: coherent—4 units.)	2.51
KHD (Asia power)-2	6.375 MW × 4	Thermal (diesel: coherent—4 units.)	2.51
Puttalam Coal	300 MW	Thermal (steam)	4
Upper kotmale	75 MW	Hydro	3.47
Heledanavi-1	17 MW × 3	Thermal (diesel: coherent—3 units.)	2.63
Heledanavi-2	17 MW × 3	Thermal (diesel: coherent—3 units.)	2.63
Kotmale-3	67 MW	Hydro	3.2
Polpitiya-1	37.5 MW	Hydro	2.84
Polpitiya-2	37.5 MW	Hydro	2.84
Canyon-1	30 MW	Hydro	3.8
Canyon-2	30 MW	Hydro	3.8
Kukule-1	35 MW	Hydro	3.0
Kukule-2	35 MW	Hydro	3.0
New Laxapana-1	50 MW	Hydro	3.3
New Laxapana-2	50 MW	Hydro	3.3
Victoria-1	70 MW	Hydro	3.45
Victoria-2	70 MW	Hydro	3.45
Victoria-3	70 MW	Hydro	3.45
Ukuwela	20 MW	Hydro	2.72
Randenigala	61 MW	Hydro	3.65
Wimalasurendra	25 MW	Hydro	3.2
Laxapana-1	8.33 MW	Hydro (coherent—3 units.)	3.17
Laxapana-2	12.5 MW	Hydro (coherent—2 units.)	2.45
Samanalawewa-1	60 MW	Hydro	3.4

(continued)

Table 3.11 (continued)

Station/generator	Unit capacity (MW)	Generator type	Inertia constant (H s)
Samanalawewa-2	60 MW	Hydro	3.4
Bowatenna	40 MW	Hydro	3.0
Rantembe-1	25 MW	Hydro	2.62
Embilipitiya	100 MW	Thermal (diesel)	3.62

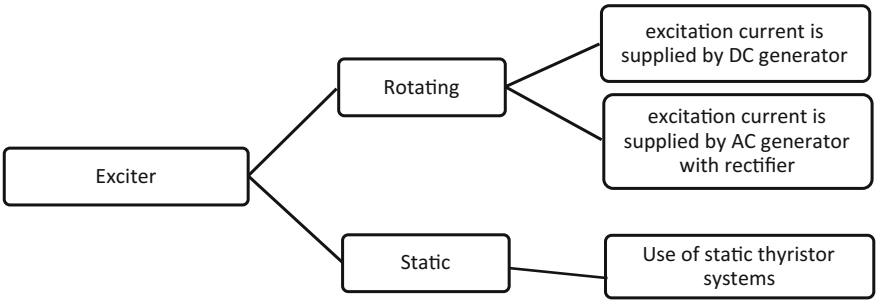


Fig. 3.16 Basic types of exciters

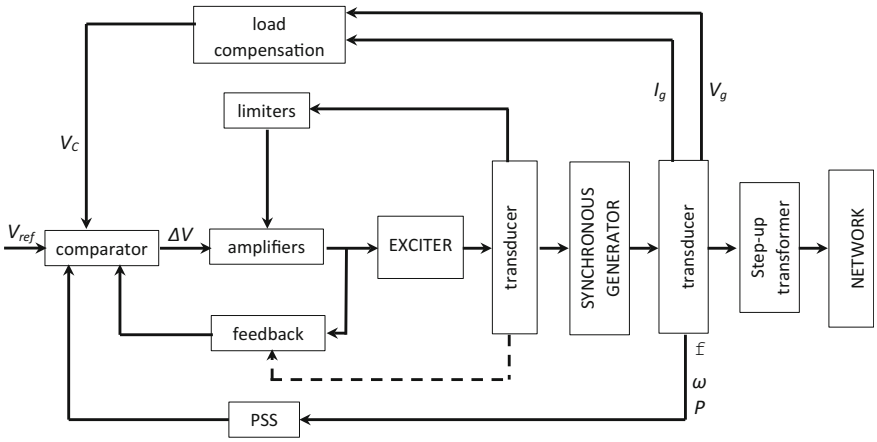


Fig. 3.17 Block diagram of the excitation and AVR system. PSS, power system stabilizer, [6]

$$\text{Base angular frequency} = 2\pi f = 2 \times \pi \times 50\,\text{rad/s} = 314.16\,\text{rad/s}$$

Finding inertia constant:

With reference to [11], the inertia constants H of synchronous generators (hydro generator and steam turbine generator) can be briefed as shown in Table 3.9. Based

on these data as well as the data obtained from the CEB (shown in Table 3.10), the inertia constants used for the simulation were calculated.

Where range is given, the 1st figure applies to the smaller MVA.

$$\begin{aligned}\text{Inertia constant (calculated) of Kotmale3 generator} &= \left(\frac{3.17-3.45}{8.3-71}\right)\times(P-8.3)+3.17\\ &= \left(\frac{3.17-3.45}{8.3-71}\right)\times(67-8.3)+3.17\\ &= 3.432\,\text{s}\end{aligned}$$

$$\text{But the experimental value} = 3.432\,\text{s}$$

$$\begin{aligned}\text{Hence, the average} &= (3.432+3.02)/2\\ &= 6.45/2\,\text{s}\\ &\sim 3.2\,\text{s}\end{aligned}$$

The inertia constants used for the corresponding units that were committed at the instant of the load-flow concerned are given in Table 3.11. Even though it has been assumed that all thermal power generators in the power system are steam-turbine power plants, in calculating inertia constants corresponding to gas-turbine/diesel power plants, they were calculated considering them as non-condensing turbine generators with inertias in the range of 4–3 s. Because of the doubts with the accuracy of experimental values obtained, in certain cases approximated values were used.

3.2.5 *Exciters*

The generator excitation system consists of an exciter and an AVR and is necessary to supply the generator with DC field current as shown in Fig. 3.16. The AVR regulates the generator terminal voltage by controlling the amount of current

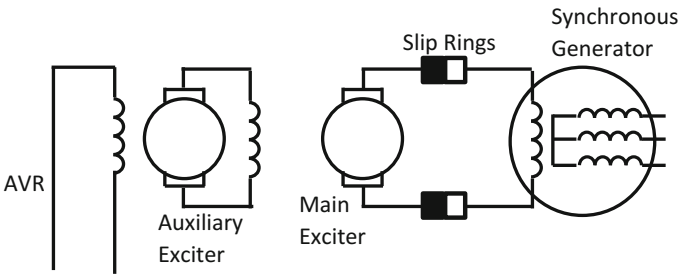


Fig. 3.18 Cascaded DC generators. AVR, automatic voltage regulator [6]

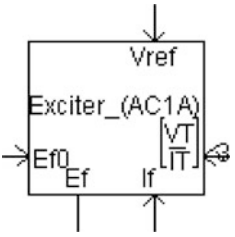


Fig. 3.19 Exciter model 'AC1A' which was used for the simulation

supplied to the generator field winding by the exciter. Generally the exciters can be classified as shown in Fig. 3.16.

Figure 3.17 demonstrates how the AVR subsystem operates [6]. The generator terminal voltage, V_g is measured and compensated for the load current I_g and compared with the desired reference voltage to get the voltage error ΔV. This error is the amplified and used to alter the exciter output, and consequently the generator field current, so that the voltage error is eliminated. This represents a closed loop

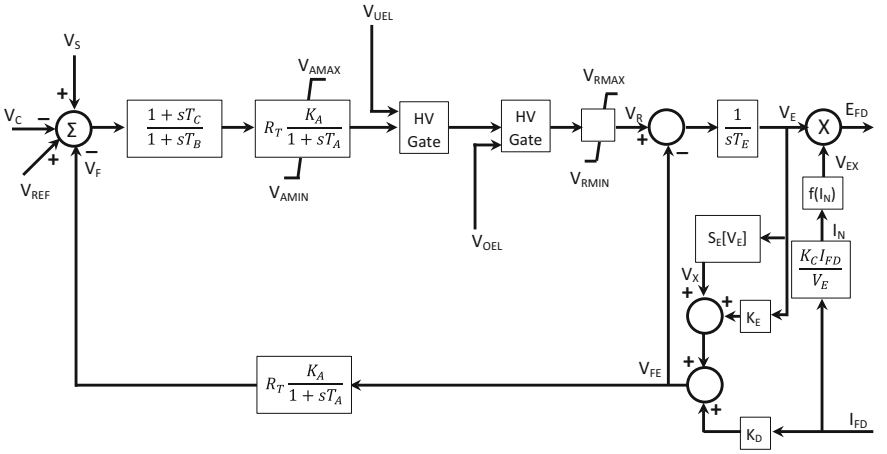


Fig. 3.20 IEEE Alternator Supplied Rectifier Excitation System #1 (AC1A) as in PSCAD. Where, Vc—output of terminal voltage and load compensation elements (pu); V_{REF}—voltage regulator reference (determined to satisfy initial conditions) (pu); V_S—combined power system stabilizer and possibly discontinuous control output after any limits or switching, as assumed with terminal voltage and reference signals (pu); V_P—excitation system stabilizer output [pu]; V_{AMAX}, V_{AMIN}—maximum and minimum regulator output limits [pu]; V_R—voltage regulator output (pu); V_{FE}—signal proportional to exciter field current; T_E—exciter time constant, integration rate associated with exciter control (s); V_E—exciter voltage back of commutating reactance (pu); K_E—exciter constant related to self-excited field (pu); K_D—demagnetizing factor, a function value at the corresponding exciter voltage, VE, back of commutation reactance (pu); K_F—excitation control system stabilizer gain [pu]; V_X—signal proportional to exciter saturation (pu); S_E [V_E]—exciter saturation function value at the corresponding exciter voltage, VE, back of commutation reactance (pu); K_C—rectifier loading factor proportional to commutation reactance (pu); I_{FD}—synchronous machine field current (pu); I_N—normalized exciter load current (pu); F_{EX}—rectifier loading factor, a function of I_N (pu); E_{FD}—exciter output voltage (pu)

control system. The regulation process is stabilized using a negative feedback loop taken directly from either the amplifier or the exciter.

Exciter Types Used in Modeling the Power System

DC generators usually have relatively low power ratings; they are cascaded to obtain the necessary output as shown in Fig. 3.18.

Because of commutation problems with DC generators, this type of exciter cannot be used for large generators which require large excitation currents (usually power rating of the exciter is in the range 0.2–0.8% of the generator's megawatt rating) [6]. Hence, in this power system simulation, for all generators, exciter model AC1A (Figs. 3.14 and 3.19), which is an ac-exciter was used.

Simplified Schematic Diagram and Corresponding Control System

Figure 3.19 shows the exciter model 'AC1A' which was used for the simulation. The control system of the 'IEEE Alternator Supplied Rectifier Excitation System #1 (AC1A)' as given in PSCAD is shown in Fig. 3.20. Further, with reference to [6], this can be considered as 'the excitation system with AC alternator and the uncontrolled rectifier' given in Fig. 3.21. Referring to Chap. 11 of [6], the corresponding values for the parameters of the AC exciter were decided. There it has been justifies that the same DC-exciter parameters could be used for the above AC-exciter system. Therefore the corresponding parameter values were decided accordingly.

The excitation system is stabilized by the feedback loop with transfer function,

$$K_G(s) = \frac{K_F s}{(1 + T_F s)}$$

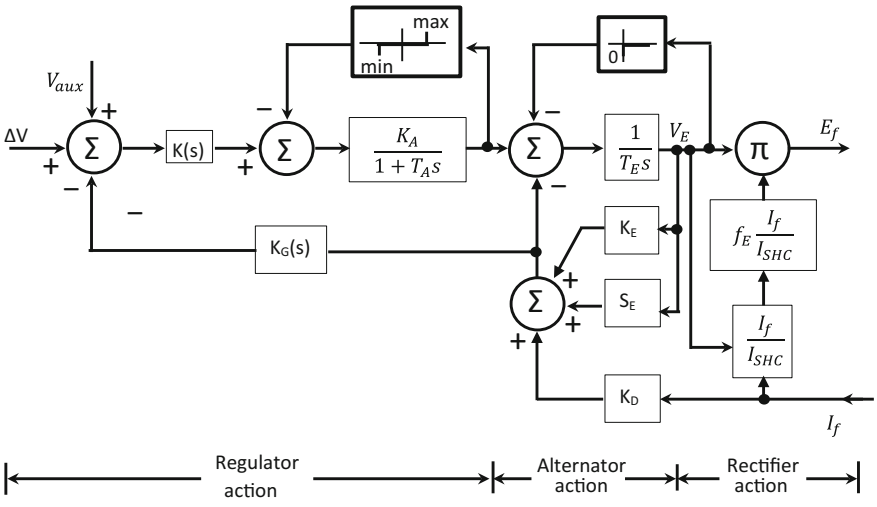


Fig. 3.21 The excitation system with AC alternator and the uncontrolled rectifier [6]

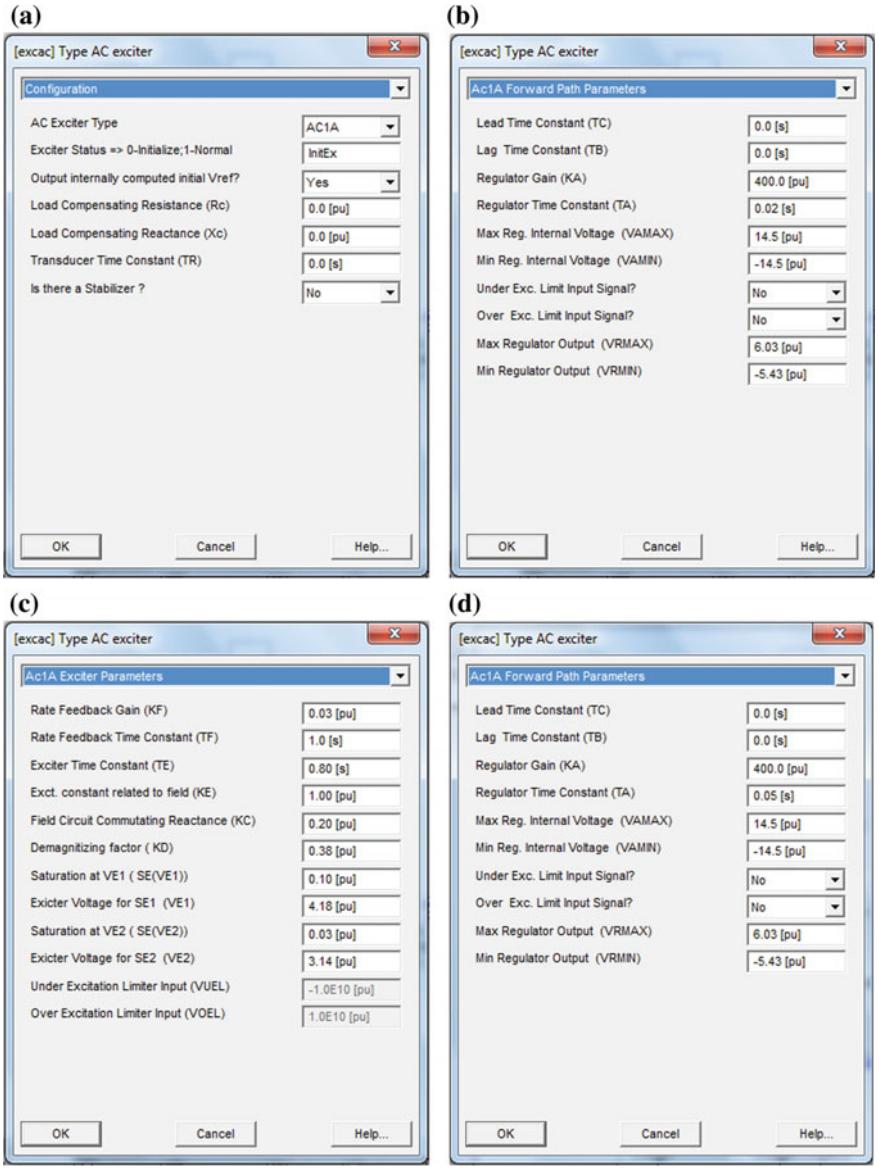


Fig. 3.22 Values used with PSCAD windows in configuring Ac1A exciters; **a** configuration, **b** hydro-gen.'s forward path parameters, **c** hydro-gen.'s exciter parameters, **d** steam turbine gen.'s forward path parameters, **e** steam turbine gen.'s exciter parameters

Alternatively the excitation system could be stabilized by supplying this block directly from the output of the voltage regulator or from the excitation voltage E_f.

In the Fig. 3.21, the feedback stabilization is done by an additional block with the transfer function K(s) in the forward path preceding the regulator block. Both

(e)

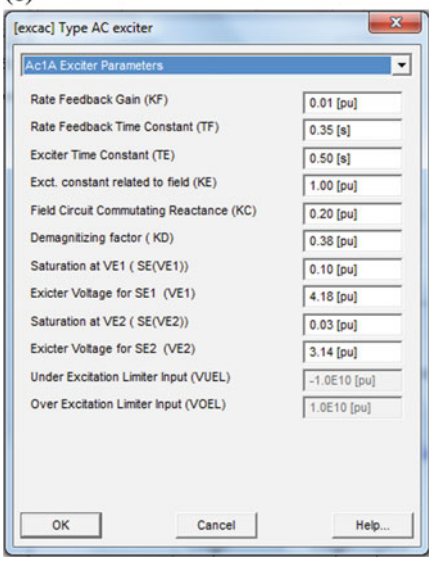


Fig. 3.22 (continued)

$K_G(s)$ and K(s) depend on the specific excitation system and can be implemented by either analogue or digital techniques. K(s) can be given as the transfer function,

$$K(s) = \frac{(1 + T_C s)}{(1 + T_B s)}$$

This can be simplified by neglecting T_C and T_B variables and thereby K(s) = 1.

With reference to [6] page 465, A (separately excited) DC exciter usually operates so that the parameter K_E = 0.8–0.95. This can be approximated to K_E = 1. The time constant, T_E < 1 s,

∴ $T_E \approx 0.5$ s. Typical values for time constant T_A = 0.05–0.2 s and gain K_A = 20–400. The high regulator gain is necessary to ensure small voltage regulation of the order of 0.5%. Although this high gain ensures low steady-state error, when coupled with the length of the time constants the transient performance of the exciter is unsatisfactory. To achieve an acceptable transient performance the system must be stabilized in some way that reduces the transient (high-frequency) gain. This is achieved by a feedback stabilization signal represented by the first-order differentiating element with gain K_F and time constant T_F. Typical values of the parameters in this element are T_F = 0.35–1 s and K_F = 0.01–0.1.

Sample Calculation

Referring to Sect. 3.2.6.2—Simplified schematic diagram and corresponding control system, Ac1A forward path parameters and exciter parameters were set. As given in Fig. 3.22, most of the parameters in control systems of 'hydro-power

generator exciter' and 'steam turbine generator exciter' are common to both, but few are different from each other.

Values Used with PSCAD Window

Figures *a, b, c, d* and *e* of Fig. 3.22 corresponds to Configuration and respective Ac1A Forward Path Parameters and Exciter Parameters of Hydro-power generator exciter and Steam turbine generator exciter.

3.2.6 Turbines

In a power system, synchronous generators are usually driven by steam turbines or gas turbines or hydro-turbines. There are two types of prime movers used for large-scale power generation. They are hydraulic (hydro) turbines, and steam turbines. The hydraulic turbine converts hydraulic energy into rotating kinetic energy of the prime mover, the steam turbine converts steam thermal energy into rotating kinetic energy of the prime mover; which is then converted into electric power by the generator [6, 40].

Turbine Types Used in Modeling the Power System
As stated in previous chapters all thermal power plants in the power system were assumed as steam turbine power plants. Therefore,

- steam turbine and
- hydro turbine

are the two types of turbines considered in this power system modelling.

Steam Turbines
In coal-burn or oil-burn power plants, the energy contained in the fuel is used to produce high-pressure, high-temperature steam in the boiler. The energy in the steam is then converted to mechanical energy in axial flow steam turbines. Each turbine consists of a number of stationary and rotating blades concentrated into groups/stages. Typically a complete steam turbine can be divided into three or more stages, with each turbine stage being connected in *tandem* on a common shaft. Dividing the turbine into stages in this way allows the steam to be reheated between stages to increase its enthalpy and consequently increase the overall efficiency of the steam cycle. Modern coal-fired steam turbines have thermal efficiency reaching 45% [6, 11]. Steam turbines can be classified according to the number of re-heat units it has.

- Non-reheat systems
- Single reheat systems
- Double reheat systems

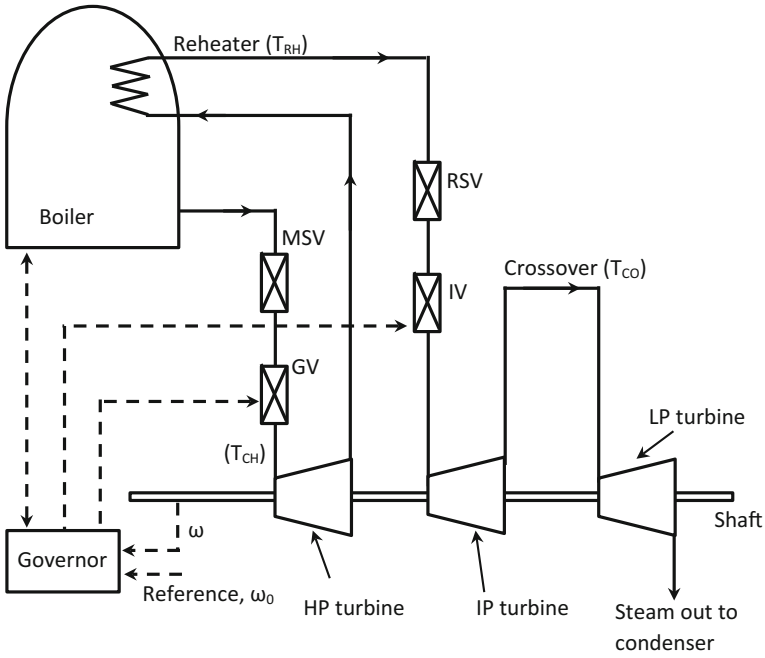


Fig. 3.23 Configuration of a tandem compound single-reheat steam turbine, [6]

Simplified Schematic Diagram and Corresponding Control System of Steam Turbine

A steam turbine configuration with a single tandem reheat arrangement is shown in Fig. 3.23. Generally a turbine has three stages in three sections,

- High pressure (HP)
- Intermediate pressure (IP)
- Low pressure (LP)

Steam from the boiler enters the steam chest and flows through the main emergency stop valve (MSV) and governor control valve (GV) to the HP turbine. In this HP turbine the steam is partially expanded (i.e. pressure is slightly reduced). Then it is directed back to the boiler to be reheated in the heat-exchanger to increase its enthalpy. The steam then flows through the reheat emergency stop valve (RSV) and the intercept control valve (IV) to the IP turbine where it is again expanded and made to do work. This IV is in use only for rapid control of turbine mechanical power during an over-speed,

- Ahead of the re-heater
- Controls steam flow to IP and LP sections.

On leaving the IP stage the steam flows through the crossover piping for final expansion in the LP turbine. Finally the steam flows to the condenser to complete

the cycle. Typically the individual turbine stages contribute to the total turbine torque in the ratio 30% (HP) : 40% (IP) : 30% (LP).

The steam flow in the turbine is controlled by the governing system (GOV). When the generator is synchronized the emergency stop valves are kept fully open and the turbine speed and power regulated by controlling the position of the GV and the IV. The speed signal to the governor is provided by the speed measuring device (SD). The main amplifier of the governing system and the valve mover is an oil servomotor controlled by the pilot valve. When the generator is synchronized the emergency stop valves are only used to stop the generator under emergency conditions, although they are often used to control the initial start-up of the turbine [6, 11].

Due to a change in GV opening, steam flow to the turbines changes. This involves some time constant, T_{CH} (∴charging time of steam chest and inlet piping to the HP section). T_{CH} = 0.2–0.3 s. Steam flow in IP and LP sections can be changed only with the buildup of pressure in the reheat volume. T_{RH} = 5–10 s. The steam flow from IP to LP through cross-over involves a time constant, T_{CO}. $T_{CO} \backsimeq 0.5$ s.

Figure 3.24 shows a block diagram representation of a control system of a steam turbine as per [11]. This model accounts for the effects of inlet steam chest, re-heater, and the nonlinear characteristics of the control and intercept valves. The re-heater representation differs from the representation of steam chest and LP inlet crossover piping. This is to allow computation of re-heater pressure to account for the effects of intercept valve actuation. 'Base power equal to the maximum turbine power at rated main steam pressure with the control valves fully open' can be considered as a convenient per unit system. In this system, CV position is 1.0 pu when fully open. Then,

$$F_{HP} + F_{IP} + F_{LP} = 1.0\,\text{pu}$$

Figure 3.24 shows a single reheat tandem-compound steam turbine model with its (a) turbine configuration and (b) block diagram representation [11].

Comparing the control systems given in PSCAD (Fig. 3.25) and [11] (Fig. 3.26) for a steam turbine, the corresponding time constants and gains were identified. The typical values given in [11] were used in configuring the steam turbine in the power system simulation model (Table 3.14).

It has been assumed that no cross over piping is available for the turbine system. Hence there is no T_{CO}. ∴ $T_{CO1} = T_{CO2} = 0.0$ s. A set of typical values as given in [11] are appeared in Table 3.12. Their ranges of existence were discussed in detail in this chapter previously.

Referring to the typical values given in [11] and Table 3.12, values for the parameters in the control system of 'Generic turbine Model including intercept valve effect as given in PSCAD' were decided. Corresponding values are appeared in Table 3.13. The same turbine model configuration was used for all thermal power plants in the PSCAD simulation model (of the power system of Sri Lanka) (Table 3.14).

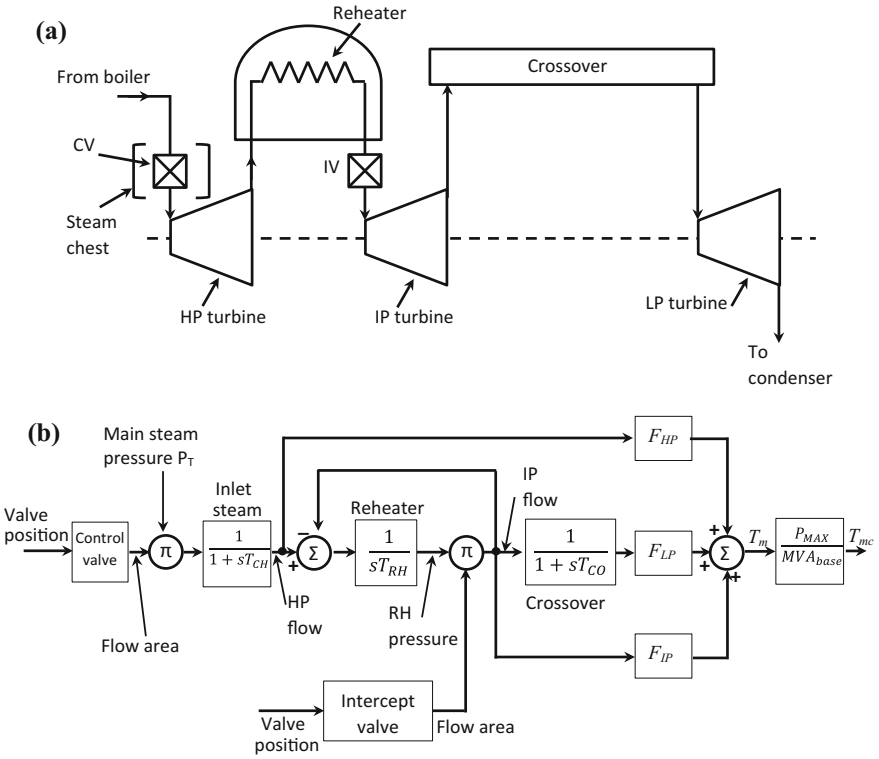


Fig. 3.24 Control system of a single reheat tandem compound steam turbine as per [11]; **a** turbine configuration; **b** block diagram representation. Where, T_{CH} = time constant of main inlet volumes and steam chest; T_{RH} = time constant of re-heater; T_{CO} = time constant of crossover piping and LP inlet volumes; T_m = total turbine torque in per unit of maximum turbine power; T_{mc} = total turbine mechanical torque in per unit of common MVA base; P_{MAX} = maximum turbine power in MW; F_{HP}, F_{IP}, F_{LP} = fraction of total turbine power generated by HP, IP, LP sections, respectively; MVA_{base} = common MVA base

Values Used with PSCAD Window

Figures *a, b, c, d* and *e* of Fig. 3.27 corresponds to actual parameter values used in 'Generic Turbine Model including Intercept Valve effect (TUR2)' Configuration, Hp Turbine: Contributions, Lp Turbine: Contributions, Time Constants and Intercept Valve.

This Steam_Tur_2 models an IEEE type thermal turbine and Fig. 3.28 shows that thermal (steam) turbine model—Generic turbine model including intercept valve effect (TUR2), which was used for the simulation.

Hydro Turbines

For hydro power plants the hydro turbine model 'Non-elastic water column without surge tank (TUR1)' was used. Figure 3.29 shows the above model as it appears in the PSCAD simulation model.

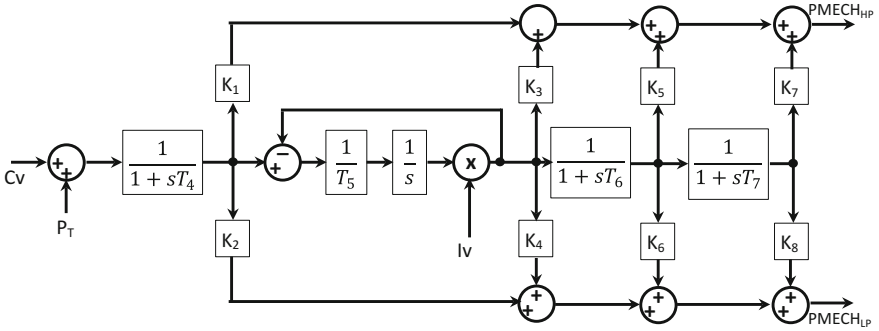


Fig. 3.25 Generic turbine model including intercept valve effect as given in PSCAD. Where, Cv = Control valve flow area (p.u.), Iv = Intercept valve flow area (p.u.), K = K fraction (p.u.), T_4 = Steam chest time constant (s), T_5 = Reheater time constant (s), T_6 = Reheater/ cross-over time constant (s), T_7 = Cross-over time constant (s), $PMECH_{HP}$ = HP turbine power output (p.u.), $PMECH_{LP}$ = LP turbine power output (p.u.)

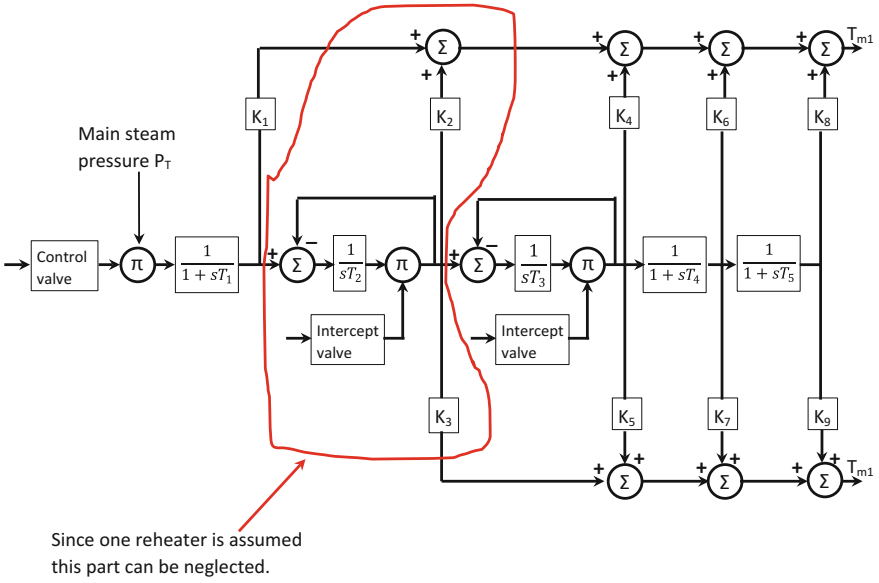


Fig. 3.26 Generic model for steam turbines [11]

The corresponding block diagram of the control system of the 'Non-elastic water column without surge tank (TUR1)' is appeared in Fig. 3.30.

Values Used with PSCAD Window

In Fig. 3.31 the parameter values with PSCAD configuration is given.

Table 3.12 Typical values in Fig. 3.26 corresponding to Fig. 3.25

Gain, K or Time-constant, T in Fig. 3.26	Corresponding values in Fig. 3.25
T_1	T_{CH}
T_2	neglected
T_3	T_{RH}
T_4	T_{CO1} (− neglected)
T_5	T_{CO2} (− neglected)
K_1	$=F_{HP}$
K_2	=0
K_3	=0
K_4	=0
K_5	F_{IP}
K_6	F_{LP1}
K_7	=0
K_8	=0
K_9	F_{LP2}

Table 3.13 Comparison of time constants and gains in Fig. 3.24 (as per [3]-Kundur's) and Fig. 3.25 (PSCAD)

Kundur's	PSCAD
T_1	$T_4 = T_{CH}$
T_3	$T_5 = T_{RH}$
T_4	$T_6 = T_{CO1}$
T_5	$T_7 = T_{CO2}$
K_1	$K_1 = F_{HP}$
K_4	K_3
K_5	$K_4 = F_{IP}$
K_6	$K_5 = F_{LP1}$
K_7	K_6
K_8	K_7
K_9	$K_8 = F_{LP2}$

Table 3.14 Values used for the parameters in the control system of 'Generic turbine Model including intercept valve effect as given in PSCAD'

PSCAD	Values as per the Fig. 3.27
T_4	T_{CH} = 0.3 s
T_5	T_{RH} = 5.0 s
T_6	T_{CO1} = 0.0 s
T_7	T_{CO2} = 0.0 s
K_1	F_{HP} = 0.297
K_2	=0.001
K_3	=0.001
K_4	F_{IP} = 0.697
K_5	F_{LP1} = 0.001
K_6	=0.001
K_7	=0.001
K_8	F_{LP2} = 0

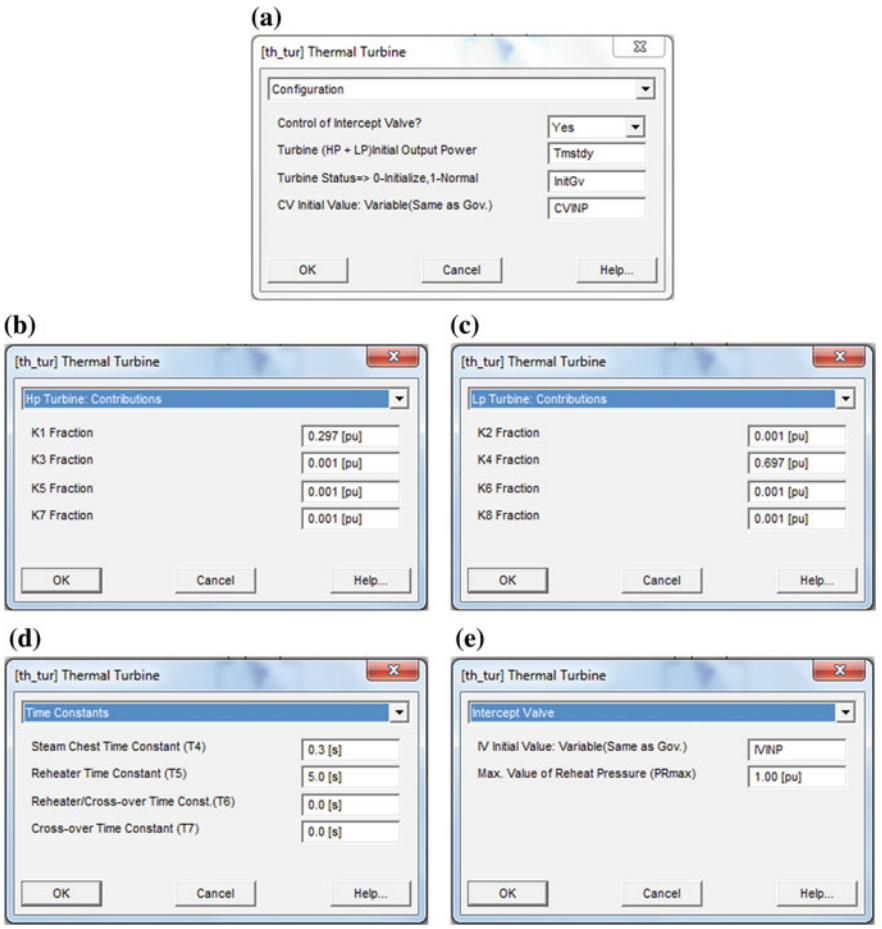


Fig. 3.27 Values used with PSCAD windows in configuring Steam_Tur_2 (Generic turbine model including intercept valve effect); **a** configuration, **b** Hp turbine: contributions, **c** Lp turbine: contributions, **d** time constants, **e** intercept valve

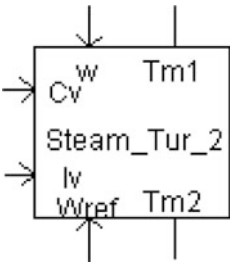


Fig. 3.28 'Steam_Tur_2' Generic turbine model including intercept valve effect (TUR2) used for the power system simulation

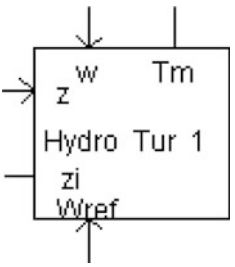


Fig. 3.29 Hydro turbine model used in power system simulation—'Non-elastic water column without surge tank (TUR1)'

3.2.7 Governors

Each turbine is equipped with a governing system (Fig. 3.11) to provide a mechanism by which the turbine can be started, run up to the operating speed and operate on load with the required power output [6, 40].

Governor Types Used in Modeling the Power System
For the steam turbines and hydro turbines in the simulation,

- steam governors and
- hydro governors

were used respectively. GOV2: Mechanical-Hydraulic Controls (GE) and GOV3: Electro-Hydraulic Controls (GE) were used in simulating steam turbine governors. For the control of hydro turbines, governor models, GOV1: Mechanical-Hydraulic Controls and GOV3: Enhanced Controls for Load Rejection Studies were used. Table 3.15 gives the Governor control systems of the PSCAD simulation with corresponding turbines.

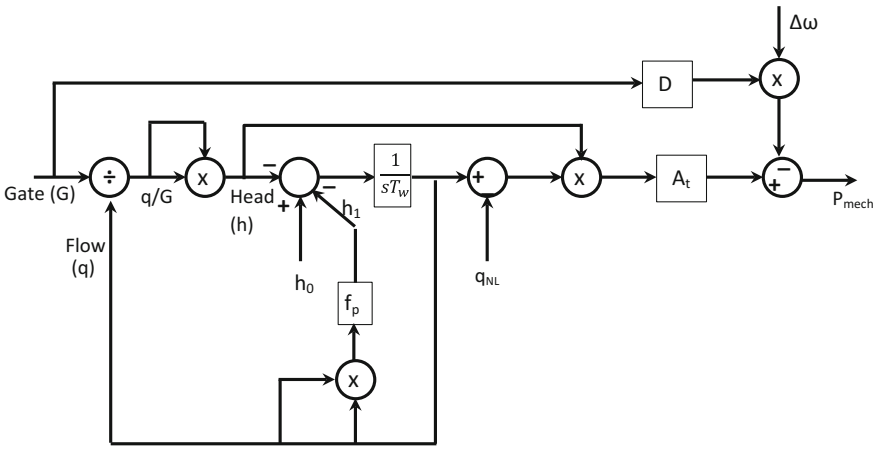


Fig. 3.30 Block diagram of control system of 'Non-elastic water column without surge tank', TUR1, PSCAD simulation model

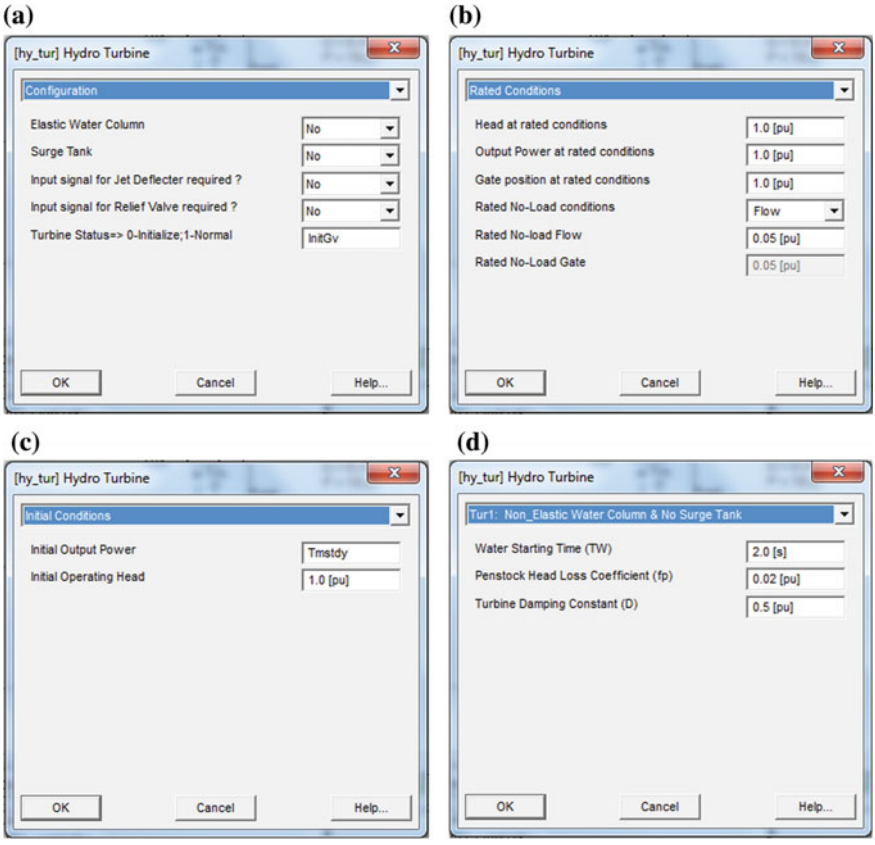


Fig. 3.31 Values used with PSCAD windows in configuring Hydro_Tur_1 (Non-elastic water column without surge tank (TUR1)); **a** configuration, **b** rated conditions, **c** initial conditions, **d** Tur: Non_Elastic Water Column and No Surge Tank

Simplified Schematic Diagram and Corresponding Control System

The Thermal (steam) governor used mostly is the Steam_Gov_2 (GOV2) but only for modelling Sapugaskanda-PP the Steam_Gov_3 (GOV3) was used. Hydro Governor-2 and 3 as appeared in PSCAD are shown in Fig. 3.32a, b respectively.

The block diagrams of the control systems of thermal governor 'Steam_Gov_2 (Mechanical-hydraulic controls (GOV2))' and 'Steam_Gov_3 (Electro-hydraulic controls (GOV3))' are appeared in Figs. 3.33 and 3.35 respectively.

As per [11], the block diagram of 'Mechanical-hydraulic controls governing system' is appeared in Fig. 3.32. By comparing the block diagrams in Figs. 3.33 and 3.34, typical values for corresponding parameters of PSCAD control system were decided. These values are appeared in Table 3.16.

Table 3.15 Governor Models of PSCAD used in controlling steam and hydro turbines of the simulation of the Power system of Sri Lanka

Thermal (steam) Governors		Hydro Governors	
Barge	GOV2	Upper Kotmale	GOV1[a]
KHD-1	GOV2	Kotmale-1	GOV3
KHD-2	GOV2	Kotmale-2	GOV3
Keravalapitiya-1	GOV2	Kotmale-3	GOV3
Keravalapitiya-2	GOV2	Kukule-1	GOV3
Keravalapitiya-3	GOV2	Kukule-2	GOV3
Kelanitissa-1 (CombinedCyclePP)	GOV2	Samanalawewa-1	GOV3
Kelanitissa-2 (CombinedCyclePP)	GOV2	Samanalawewa-2	GOV3
Sapugaskanda-1	GOV3[b]	Canyon-1	GOV3
Sapugaskanda-2	GOV3[b]	Canyon-2	GOV3
Sapugaskanda-3	GOV3[b]	New Laxapana-1	GOV3
Coal Power Plant	GOV2	New Laxapana-2	GOV3
Heladanavi-1	GOV2	Laxapana-1	GOV3
Heladanavi-2	GOV2	Laxapana-2	GOV3
Embilipitiya	GOV2	Polpitiya-1	GOV3
		Polpitiya-2	GOV3
		Victoria-1	GOV3
		Victoria-2	GOV3
		Victoria-3	GOV3
		Wimalasurendra	GOV3
		Randenigala	GOV3
		Ukuwela	GOV3
		Bowatenna	GOV3
		Rantembe	GOV3

[a]When run the simulation load flow was stopped with 'Upper Kotmale PP with GOV3.' This could be eliminated by using GOV1

[b]% over shoot of power (of SapugaskandaPP) during starting of the generator (during the transient period) could be minimized with GOV3 than with GOV2

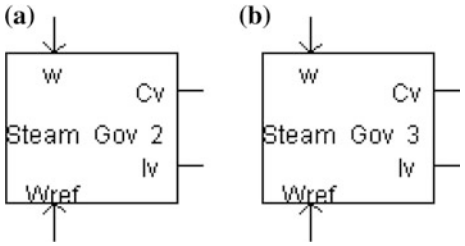


Fig. 3.32 Governor models used in PSCAD;
a Steam_Gov_2 (GOV2)
b Steam_Gov_3 (GOV3)

As per [11], the block diagram of 'Electro-hydraulic controls governing system' is appeared in Fig. 3.36. By comparing the block diagrams in Figs. 3.35 and 3.36, typical values for corresponding parameters of PSCAD control system were decided. These values are appeared in Table 3.17

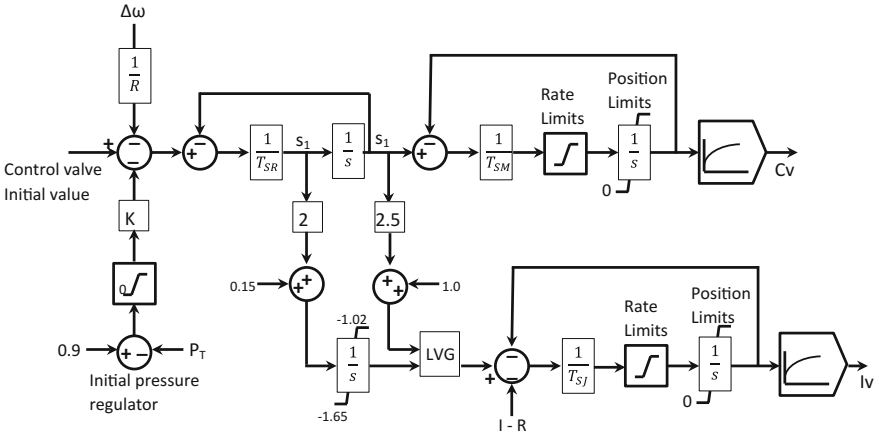


Fig. 3.33 Block diagram of control system of Steam_Gov_2 (Mechanical-hydraulic controls), PSCAD simulation model. Where, Cv—Control valve floor area (p.u.); Iv—Intercept valve floor area (p.u.); R—Permanent Droop (p.u.); T_{SJ}—IV servo time constant (s); T_{SM}—Gate servo time constant (s); T_{SR}—Speed relay lag time constant (s)

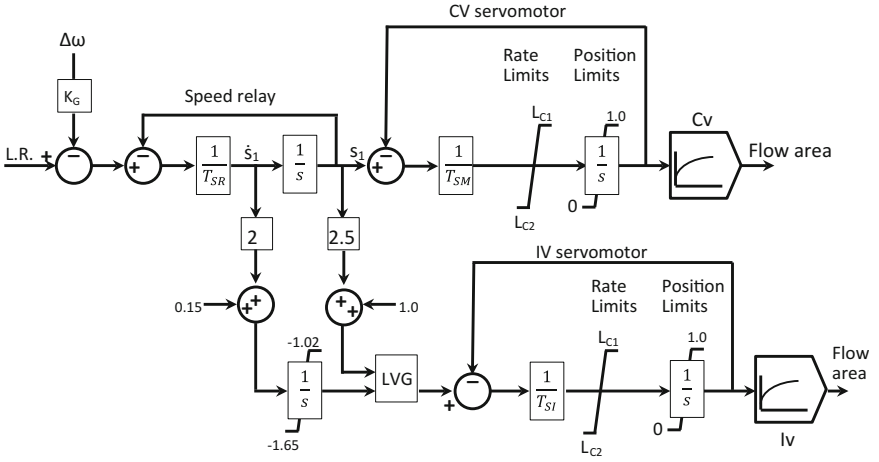


Fig. 3.34 Block diagram of 'Mechanical-hydraulic controls governing system' as per [11]

The hydro turbine governor used mostly is the Hydro_Gov_3 (GOV3) but only for modelling Upper-Kotmale-PP the Hydro_Gov_1 (GOV1) was used. Hydro Governor-1 and 3 as appeared in PSCAD are shown in Fig. 3.37a, b respectively.

The block diagrams of the control systems of hydro governor 'Hydro_Gov_1 (Mechanical-hydraulic controls (GOV1))' and 'Hydro_Gov_3 (Enhanced controls for load rejection studies (GOV3))' are appeared in Figs. 3.38 and 3.40 respectively.

Table 3.16 Typical values for the parameters in Mechanical-Hydraulic Controls (GE) (GOV2) model in PSCAD

Parameter as per [11]	Corresponding parameter in PSCAD	Typical value
K_G	1/R	20
T_{SR}	T_{SR} (Speed Relay time constant)	0.1 s
T_{SM}	T_{SM} (CV servo-motor time constant)	0.2 s
T_{SI}	T_{SJ} (IV servo-motor time constant)	0.2 s
L_{C1}	P_up	0.2 pu/s
L_{C2}	P_down	−0.5 pu/s
L_{I1}	I_up	0.2 pu/s
L_{I2}	I_down	−0.5 pu/s

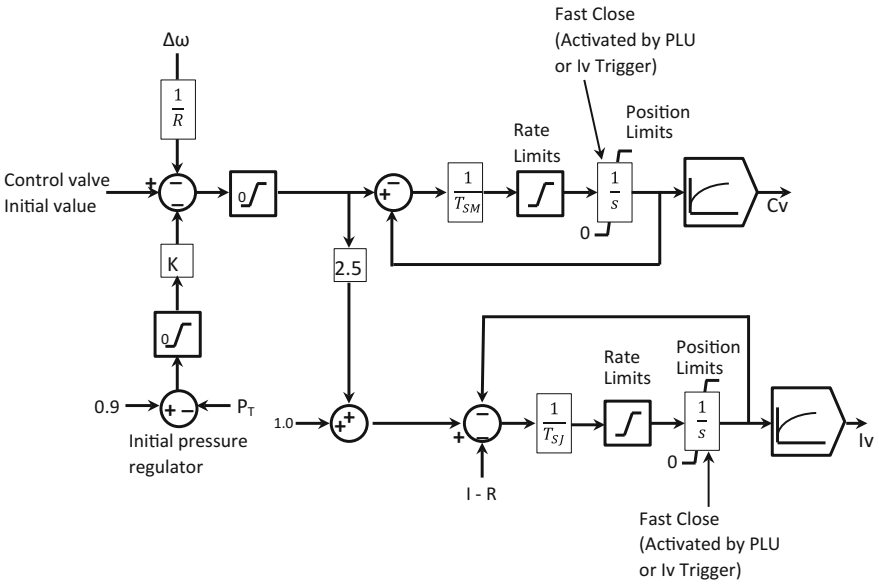


Fig. 3.35 Block diagram of control system of governor Electro-Hydraulic Controls (GOV3), as per PSCAD simulation model. Where, Cv—Control valve floor area (p.u.); Iv—Intercept Valve Flow Area (p.u.); R—Permanent Droop (p.u.); T_{SJ}—IV Servo Time Constant (p.u.); T_{SM}—Gate servo time constant [p.u.]; P_T—Steam pressure (constant)

As per [11], the block diagram of 'Electro-hydraulic controls governing system' is appeared in Fig. 3.39. By comparing the block diagrams in Figs. 3.38 and 3.39, typical values for corresponding parameters of PSCAD control system were decided. These values are appeared in Table 3.18 (Fig. 3.40).

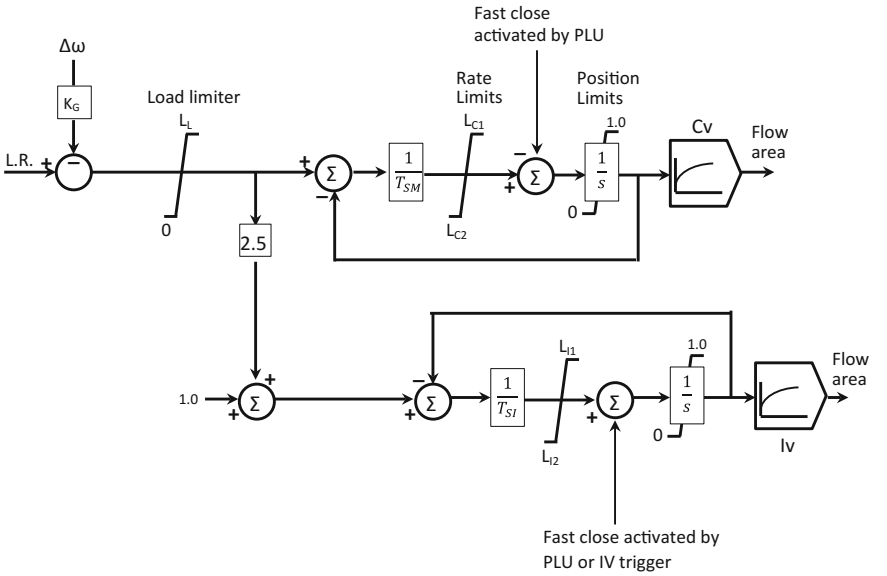


Fig. 3.36 Block diagram of 'Electro-hydraulic controls governing system' as per [11]

Table 3.17 Typical values for the parameters in Electro-Hydraulic Controls (GE) (GOV2) model of a steam turbine in PSCAD

Parameter as per [11]	Corresponding parameter in PSCAD	Typical value
K_G	1/R	20
T_{SM}	T_{SM} (CV servo-motor time constant)	0.1 s
T_{SI}	T_{SJ} (IV servo-motor time constant)	0.1 s
L_{C1}	P_up	0.1 pu/s
L_{C2}	P_down	−0.2 pu/s
L_{I1}	I_up	0.1 pu/s
L_{I2}	I_down	−0.2 pu/s

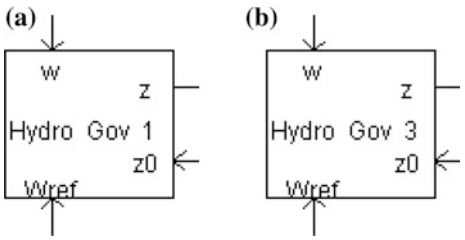


Fig. 3.37 Governor models used in PSCAD;
a Hydro_Gov_1 (GOV1)
b Hydro_Gov_3 (GOV3)

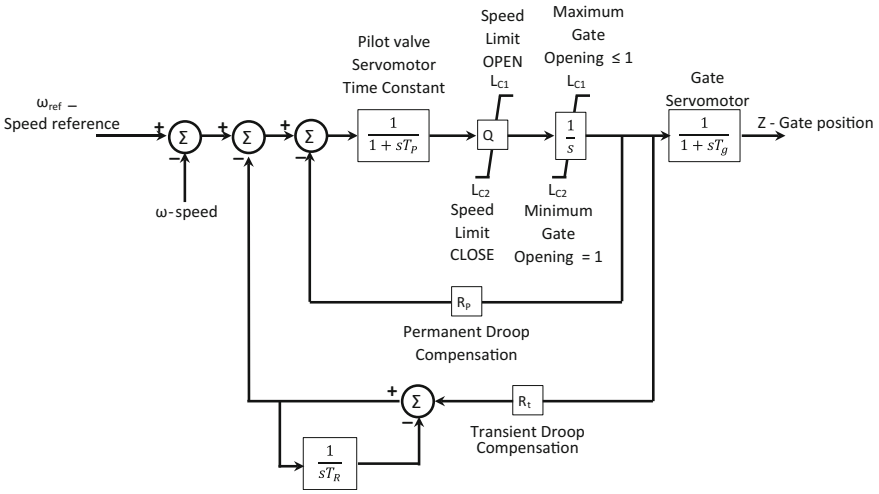


Fig. 3.38 Block diagram of the control system of hydro governor 'Hydro_Gov_1 (Mechanical-hydraulic controls (GOV1) as per PSCAD simulation model). Where, Q—servo Gain (p.u.); Rp—Permanent Droop (p.u.); Rt—Temporary Droop (p.u.); Tg—Main Servo time constant (s); Tp—Pilot valve and servo motor time constant (s); Tr—Reset or Dashpot time constant (s)

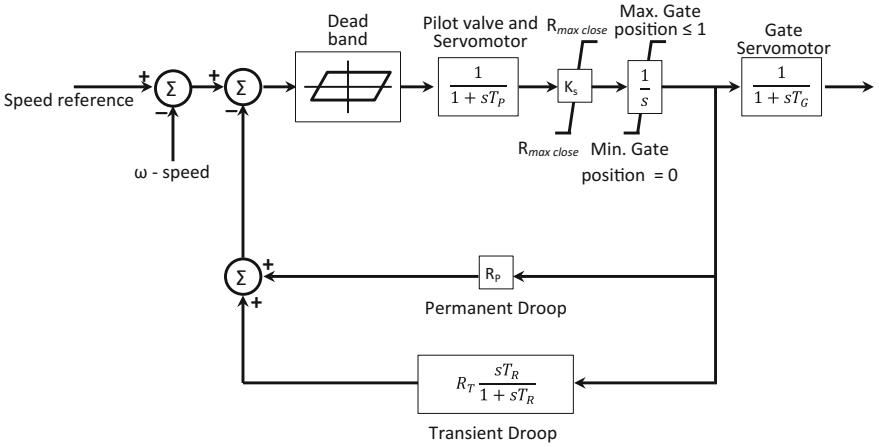


Fig. 3.39 Block diagram of governing system for hydraulic turbine as per [3]

With reference to [11], the block diagram shown in Fig. 3.39 and sample data in Table 3.19 provide values to configure the governor model 'Hydro_Gov_3 (Enhanced controls for load rejection studies (GOV3))' of PSCAD.

Values Used with PSCAD Window

Figure 3.41 shows the parameter values for thermal (steam) turbine governor types used in the simulation of power system of Sri Lanka: Steam_Governor_2 and Steam_Governor_3 with a. PSCAD configuration for Steam _Gov_2, b. PSCAD configuration for Steam _Gov_3, c. parameters of Gov2: Mechanical-Hydraulic (GE) Governor: section 1, d. parameters of Gov2: Mechanical-Hydraulic (GE) Governor: section 2, e. parameters of Gov3: Electro-Hydraulic (GE) Governor: section 1 and *f.* parameters of Gov3: Electro-Hydraulic (GE) Governor: section 2.

Figure 3.42 shows the parameter values for hydro turbine governor types used in the simulation of the power system of Sri Lanka: Hydro_Governor_1 and Hydro_Governor_3 with a. PSCAD configuration for Hydro_Gov_1, b. PSCAD configuration for Hydro_Gov_3, c. parameters common to both types, d. Gov_1 Mechanical-hydraulic governor parameters, e. Gov_3 Enhanced Governor parameters are given.

3.3 Control System of the Overall Power System

The control system of the overall power system is comprised with three modules. They are:

• LS Logic1	• Senses the frequency and voltage of the system (Biyagama Grid-substation) • Filters the frequency and voltage with 'Low pass, Chebyshev, Order 3' filter and 'Low pass, Butterworth, Order 2' filters respectively • These values are used to implement the under frequency Load Shedding scheme (without going for islanding operation)
• U_Frequency	• Senses the frequency and voltage of the system (Biyagama Grid-substation) • Filters the voltage and frequency with 'Low pass, Chebyshev, Order 3' filters • These values are used to implement the tripping action of generators during low frequencies (f < 47 Hz)
• Add_Ld	• To see the system response (frequency and voltage) after a sudden load addition in the power system (here it was performed at Biyagama grid sub-station)

3.3.1 LS Logic1 Control System Module

Two types of filters were used in measuring 'system frequency' and 'system voltage.' For the measurement of the system frequency, Low pass, Chebyshev,

Table 3.18 Typical values for the parameters in Mechanical-Hydraulic Controls (GE) (GOV1) model of a hydro turbine in PSCAD

Parameter as per [11]	Corresponding parameter in PSCAD	Typical value
T_P	T_P	0.05 s
K_S	Q	5.0
T_G	T_g	0.2 s
R_P	R_P	0.04
R_T	R_t	0.4
T_R	T_R	5.0 s
Max. gate position	Gmax	1
Min. gate position	Gmin	0
$R_{max\ open}$	MXGTOR	0.16 pu/s
$R_{max\ close}$	MXGTCR	0.16 pu/s

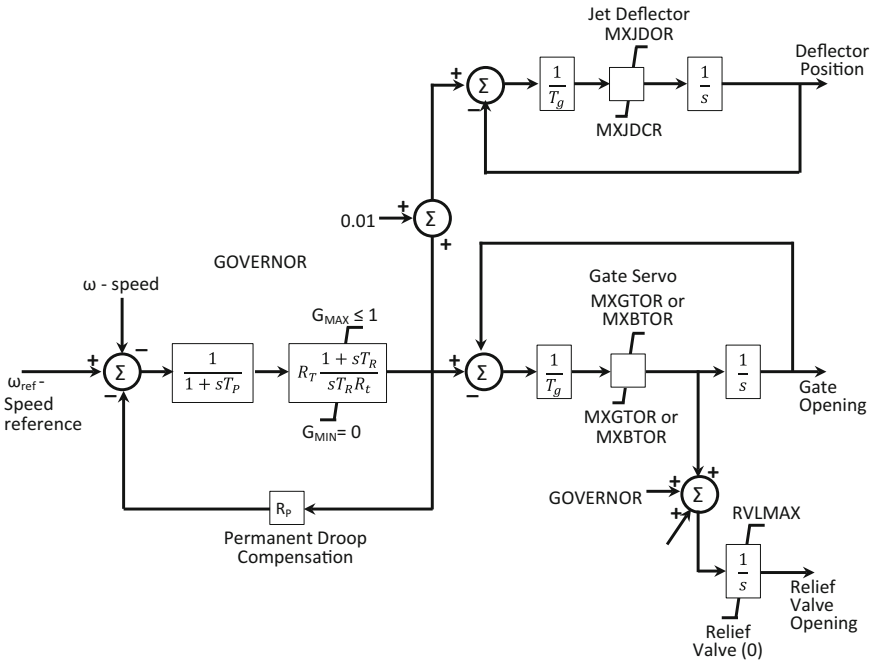


Fig. 3.40 Hydro_Gov_3 (Enhanced controls for load rejection studies (GOV3) of PSCAD). Where, G_{MAX}—Maximum Gate Position (p.u.); G_{min}—Minimum Gate Position (p.u.); MXBGCR—Maximum Gate Buffer Closing Rate (p.u./s); MXBGOR—Maximum Gate Buffer Opening Rate (p.u./s); MXGTCR—Maximum Gate Closing Rate (p.u./s); MXGTOR—Maximum Gate Opening Rate (p.u./s); MXJDCR—Maximum Jet Deflector Closing Rate (p.u./s); MXJDOR—Maximum Jet Deflector Opening Rate (p.u./s); RVLMAX—Maximum Relief Valve Position (p.u.); RVLCVR—Relief Valve Closing Rate (p.u./s); R_p—Permanent Droop (p.u.); R_t—Temporary Droop (p.u.); T_g—Main Servo Time Constant (s); T_p—Pilot valve and servo motor time constant (s); T_R—Reset or Dashpot Time Constant (s)

Table 3.19 Values used to configure the governor model 'Hydro_Gov_3 (enhanced controls for load rejection studies (GOV3))'

Parameter as per [11]	Corresponding parameter in PSCAD	Typical value
T_P	T_P	0.05 s
K_S	Q	5.0
T_G	T_g	0.2 s
R_P	R_P	0.04
R_T	R_t	0.4
T_R	T_R	5.0 s
Max. gate position	Gmax	1
Min. gate position	Gmin	0
$R_{max\ open}$	MXGTOR	0.16 pu/s
$R_{max\ close}$	MXGTCR	0.16 pu/s
	MXBGOR	
	MXBGCR	
	MXJDOR	
	MXJDCR	

order-3 filter was employed and for the measurement of the system voltage, Low pass, Butterworth, order-2 was employed.

Even though the Chebyshev has a ripple in its pass-band, it optimizes the roll-off time [47]. The main intention of the design is to suggest an under frequency load shedding scheme for the power system of Sri Lanka. So it is necessary to filter-out the required low frequencies to get the actual power system frequencies during normal and abnormal situations. The order of the Chebyshev filter was decided experimentally. It is required to maintain the system frequency at 50 Hz under steady state condition. Selection of the type of filters was done experimentally. Table 3.20 shows the 'Maximum overshoot during start of source' and 'Steady state frequency' obtained with different Chebyshev filters.

Referring to the results obtained in Table 3.20, the Chebyshev filter with 'Order = 3, Base frequency = 3 Hz, Ripple = 0.05 dB, Maximum overshoot during start of source =55.01 Hz, Steady state frequency = 50 Hz' was decided to use in detection of frequency, in modeling the control system of the power system. Figure 3.43 shows the PSCAD windows corresponding to the Chebyshev filter configuration.

Since the Butterworth filter optimizes the flatness of the passband [47], a 'Low pass, Order 2, Butterworth filter with base frequency 0.5 Hz' was used to filter-out the voltage signal for measuring purposes. Figure 3.44 shows the PSCAD windows for Butterworth filter configuration.

Figure 3.45 shows the Chebyshev and Butterworth filters used in measurement of System frequency and System voltage respectively, in the control system of the simulation model.

In-order to get the rate of change of frequency, it is possible to use the model 'derivative with a time constant' in PSCAD which is shown in Fig. 3.46. As per the PSCAD help itself says [37], 'This block is fraught with danger because of its

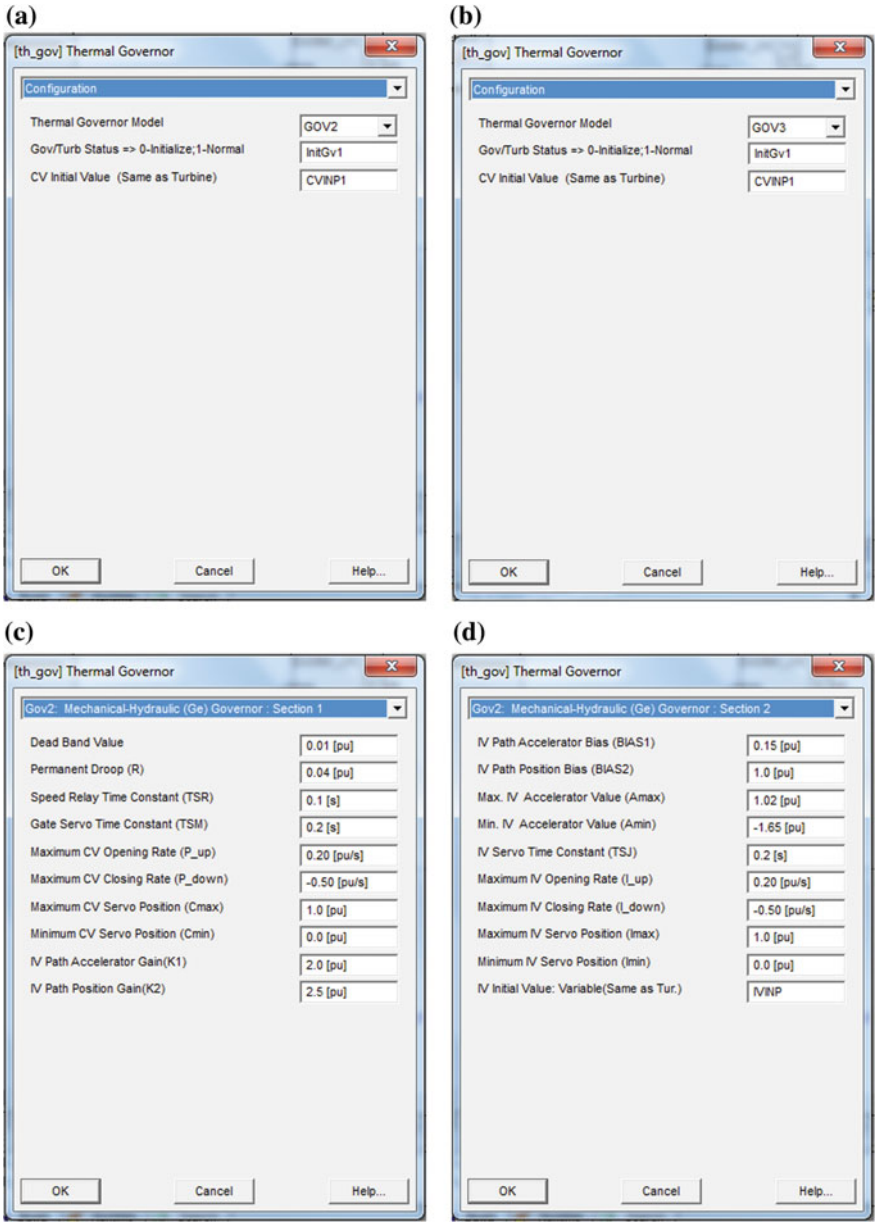


Fig. 3.41 Steam_Governor_2 and Steam_Governor_3 with **a** PSCAD configuration for Steam _Gov_2, **b** PSCAD configuration for Steam _Gov_3, **c** parameters of Gov2: Mechanical-Hydraulic (GE) Governor: section 1, **d** parameters of Gov2: Mechanical-Hydraulic (GE) Governor: section 2, **e** parameters of Gov3: Electro-Hydraulic (GE) Governor: section 1, **f** parameters of Gov3: Electro-Hydraulic (GE) Governor: section 2

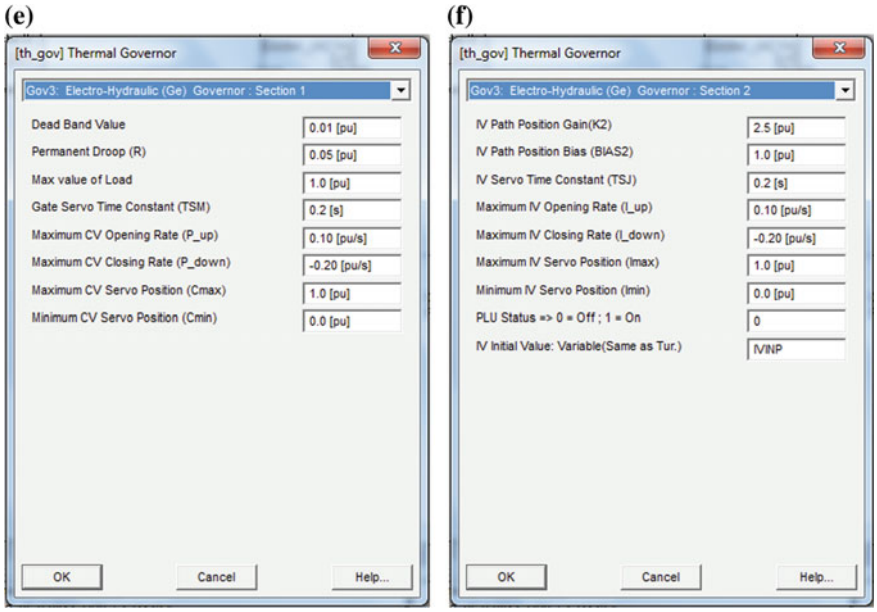


Fig. 3.41 (continued)

tendency to amplify noise. To minimize noise interference, particularly when the derivative time constant is large and the calculation step is small, it may be necessary to add a noise filter.' Further, the model time interval Δt can't be varied considering the situation to which we apply. This considers the value of the 'solution time step' which is the time step taken in PSCAD runtime to work out the whole simulation [30]. For the simulation of the power system of Sri Lanka, this 'solution time step' has been taken as 100 μs.

There for to find the rate of change of frequency (df/dt), the model shown in Fig. 3.47 has been used. It clearly demonstrates the output function given in Fig. 3.46b, [48]. In this model,

Table 3.20 Maximum overshoot during start of source and steady state frequency obtained with different Chebyshev filter settings

Details of the Chebyshev filter used					
Order	2	3	3	3	4
Base frequency (Hz)	3	3	3	3	3
Ripple (dB)	0.05	0.05	0.1	0.03	0.05
Maximum overshoot during start of source (Hz)	52.66	55.01	55.08	54.92	56.56
Steady state frequency (Hz)	49.75	50	50.01	$50.00 \leq f \geq 50.015$	49.76

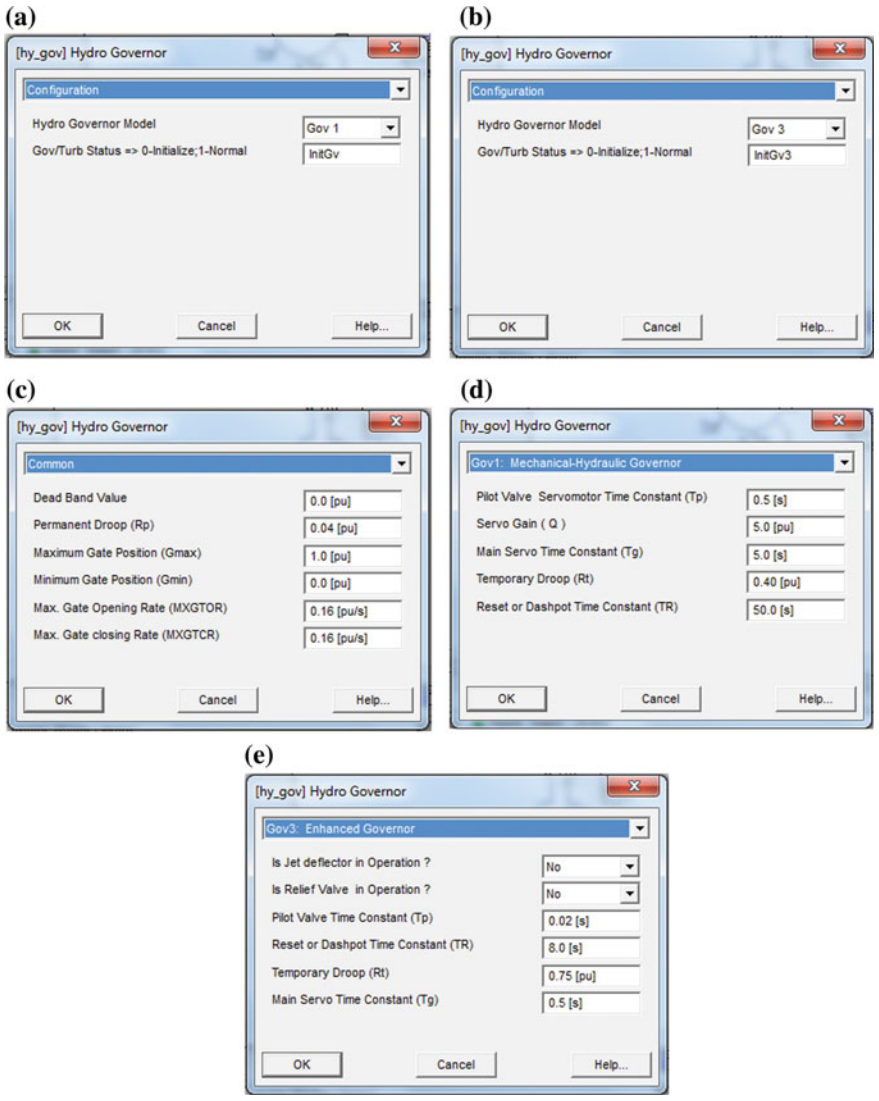


Fig. 3.42 Parameter values for hydro turbine governor types used in the simulation model a PSCAD configuration for Hydro_Gov_1, **b** PSCAD configuration for Hydro_Gov_3, **c** parameters common to both types, **d** Gov_1 Mechanical-hydraulic governor parameters, **e** Gov_3 Enhanced Governor parameters

Delay time, T	=0.01 s
Number of samples in delay T	=1
Time step of derivative, Δt	=0.01 s

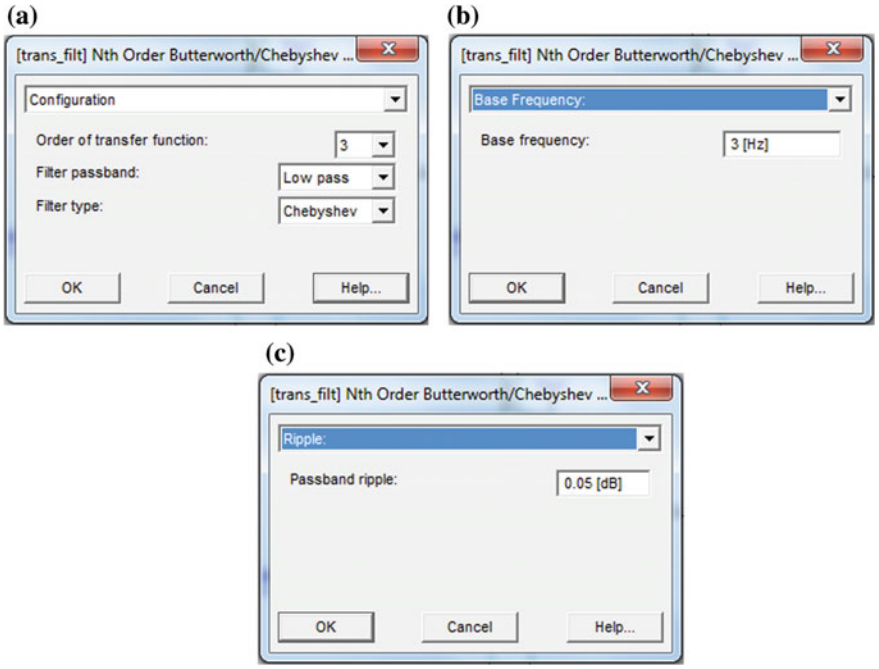


Fig. 3.43 PSCAD windows corresponding to the Chebyshev filter configuration a PSCAD configuration of Low pass, Order 3, Chebyshev filter, **b** Chebyshev filter's base frequency has been set to 3 Hz, **c** its ripple has been set to 0.05 dB

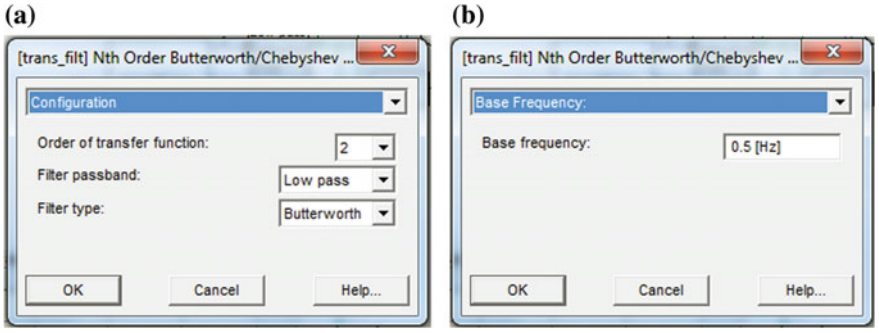


Fig. 3.44 PSCAD windows with **a** Butterworth filter configuration, **b** base frequency set to 0.5 Hz

With the available power generation, the power system can withstand a sudden generation deficit of 20 MW. At that instant the rate of change of frequency is $\backsimeq -0.03$ Hz/s. The relay settings for load shedding should be done only if the output signal $y(t) < -0.03$ Hz/s. A delay time that corresponds to relay operating time, circuit breaker operating time and an intentionally set delay time, should also be

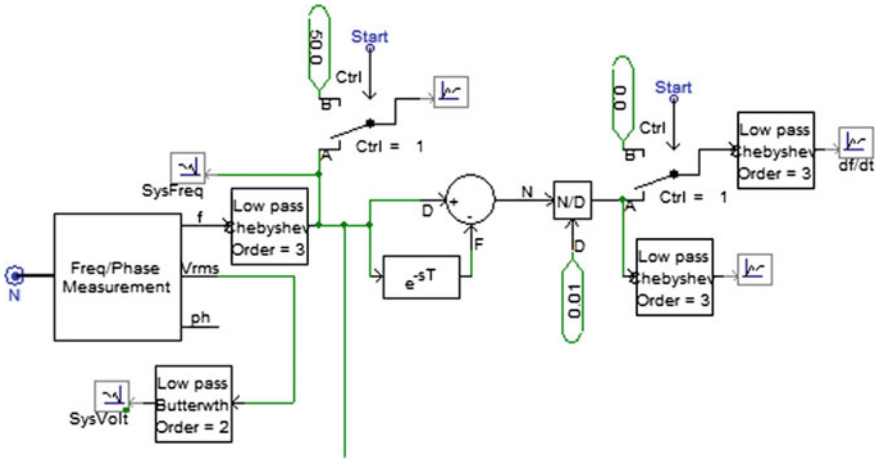


Fig. 3.45 The Chebyshev and Butterworth filters used in measurement of system frequency and system voltage respectively, in the control system of the simulation model

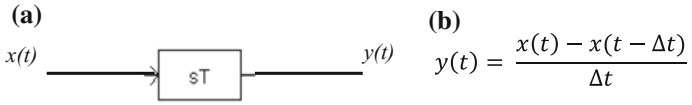


Fig. 3.46 Derivative function in PSCAD; **a** block diagram of the DERIV function, **b** the output as a function of the input

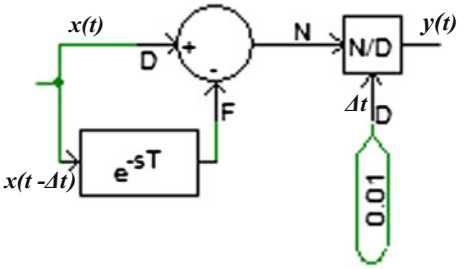


Fig. 3.47 PSCAD model used to calculate the df/dt of the power system

considered in sending a signal for the relay to operate. The PSCAD model 'sample and hold' is used to send the correct signal to the relay to get operated. This corresponds to different stages of the load shedding scheme. Figure 3.48 shows the control circuit of this operation.

Figure 3.49 demonstrates how the system responds to a sudden generation deficit of 23.18 MW (by tripping generator Wimalasurendra). The steady state frequency and voltage it achieves are 49.5 Hz and 211.81 kV respectively. With this disturbance it achieves a frequency degradation rate of −0.038 Hz/s. Victoria2

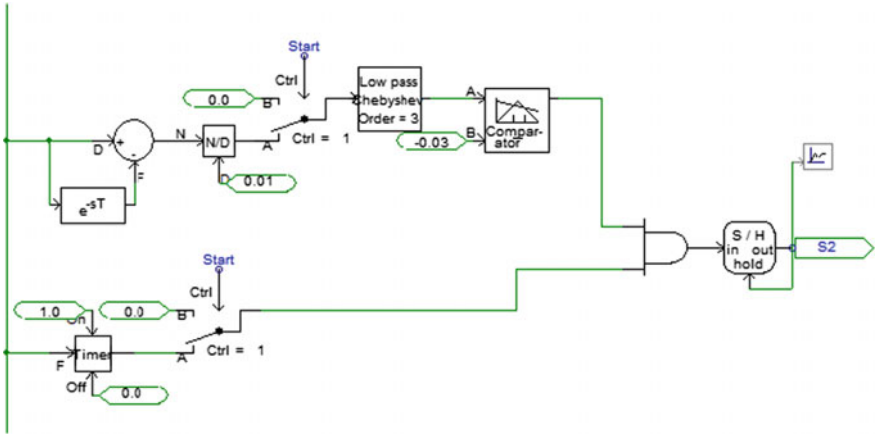


Fig. 3.48 The control circuit which senses the system frequency and to operate circuit breakers

which has been set as the isochronous governor reaches to its maximum output with this situation.

3.3.2 U_Frequency Control System Module

The same frequency measuring control techniques have been employed as discussed in Sect. 3.3.1. The purpose of the control system module is to trip off the thermal generators at 47 Hz after sensing the system frequency. Since the operation of thermal power plants are entirely not good for its performance and their operation, this action is taken place.

3.3.3 Add_Ld Control System Module

This control system module is to see the power system behavior during a sudden load addition (e.g. 5.75% of the total demand). The control logic has been done using sequencer components as shown in Fig. 3.50.

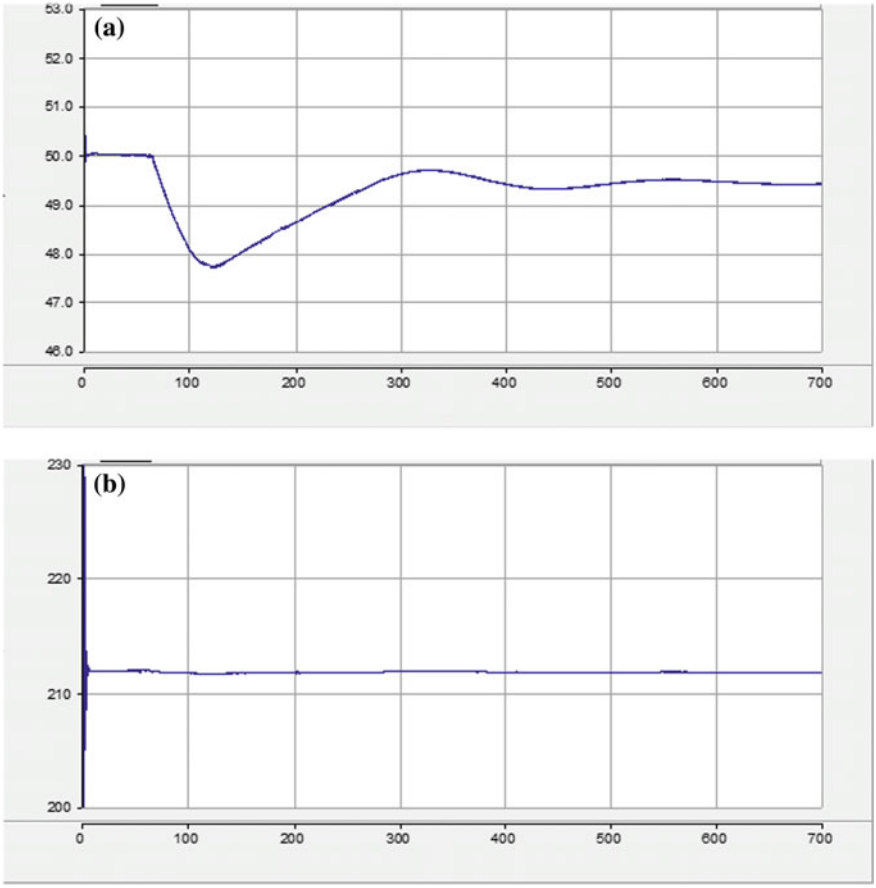


Fig. 3.49 System responses to a sudden generation deficit of 23.18 MW (by tripping generator Wimalasurendra); **a** Steady state frequency achieved = 49.5 Hz; **b** Steady state voltage achieved = 211.81 kV; **c** generation output of Victoria2 which has been set as the generator with isochronous governor; **d** rate of change of frequency in response to the disturbance—the minimum rate achieved = −0.038 Hz/s

3.4 Verifying the Simulation Model Performance

3.4.1 Steady State Operation

As reported in previous sections of this chapter, the power system simulation model was designed with PSCAD/EMTDC software. So it is very important to see its performance during normal and abnormal conditions of the power system. The load flow on the 13th March, 2013 during daytime peak demand has been simulated in the simulation model. It is very important to see how well it performs compared to the actual power system components in operation at the same instant.

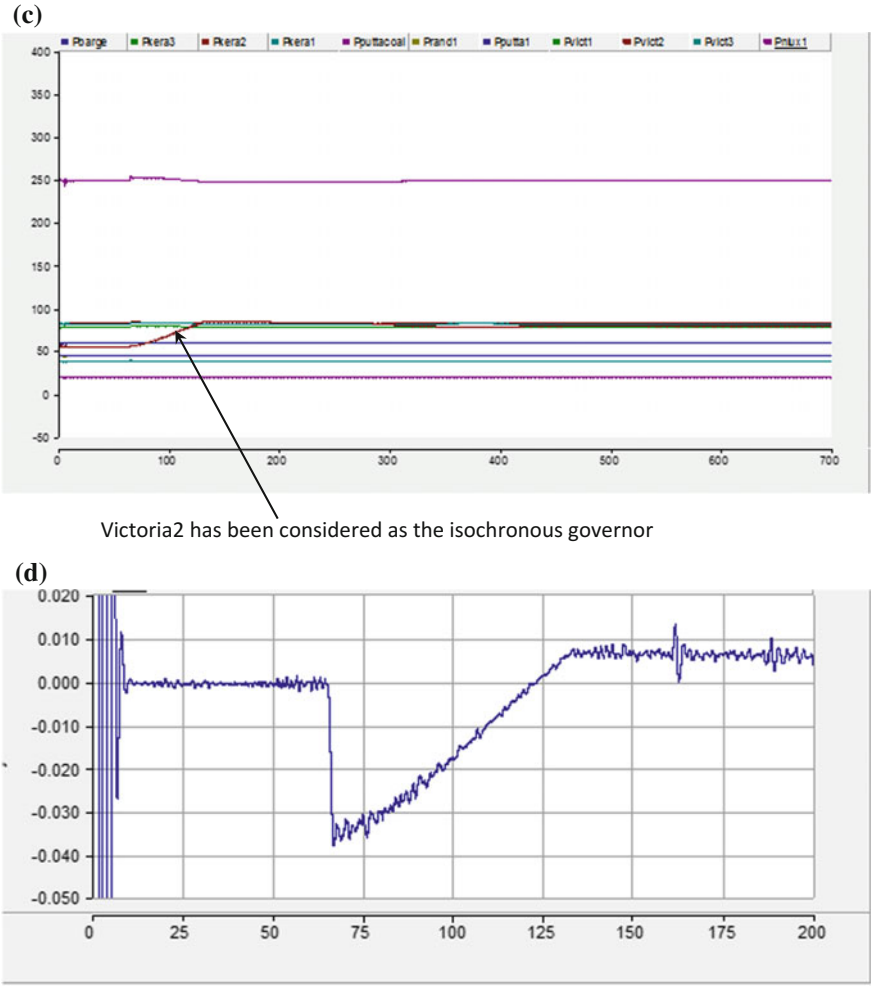


Fig. 3.49 (continued)

Corresponding actual generator outputs and simulation model's generator outputs of the load flow on the 13th March, 2013 during day time peak demand are tabulated in Table 3.21.

The voltage, frequency and some of the generator outputs are shown in the Fig. 3.51 under steady state operation. The program was run for 100 s. the lines of the power generation outputs corresponds to Coal power land (248.8 MW), Kerawlapitiya1 (82.25 MW), Kerawalapitiya3 (78.43 MW), Victoria1 and 3 (38.55 MW), Victoria2 (55.67 MW), Puttalam1 (47.58 MW), NewLaxapana1 (19.47 MW) and NewLaxapna2 (19.6 MW).

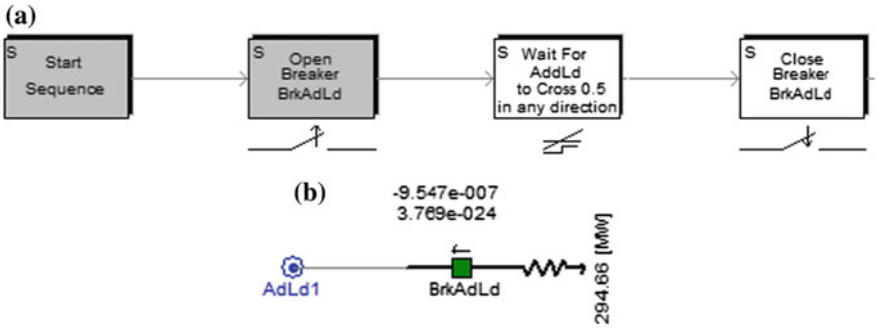


Fig. 3.50 A sudden load addition at the required instant, which was implemented using sequencer element; **a** series of sequencer elements used for the operation, **b** a resistive load was connected to the system through the breaker 'BrkAdLd'

Table 3.21 Actual generator outputs and simulation model's generator outputs corresponding to the load flow on the 13th March, 2013 during day time peak demand

Generators committed to the system	Actual generator outputs		Simulation model's generator outputs	
	Active power (MW)	Reactive power (MVAR)	Active power (MW)	Reactive power (MVAR)
Barge	60.5	35	59.7	27.62
KCCPgas	99	77	89.35	58.97
KCCPsteam	54	29.5	49.74	28.71
Sapugaskanda-1	25.5	8	17.69	16.45
Sapugaskanda-2	18	8	17.69	16.79
Sapugaskanda-3	12.5	12	35.37	29.68
Kerawalapitiya-1	90	50	82.23	10.68
Kerawalapitiya-2	90	50	82.43	43.54
Kerawalapitiya-3	90	50	78.41	31.93
KHD (Asia power)-1	17.5	2	16.95	26.12
KHD (Asia power)-2	17.5	2	16.95	26.12
Puttalam Coal	247	110.8	248.7	64.99
Upper kotmale	60	19.6	69.16	50.71
Heledanavi-1	40.15	5.9	47.58	3.4
Heledanavi-2	40.15	5.9	47.58	3.4
Kotmale-3	67	45	69.71	63.1
Polpitiya-1	32.3	17	28.03	21.89
Polpitiya-2	32	17	27.76	21.91
Canyon-1	0			
Canyon-2	0			

(continued)

Table 3.21 (continued)

Generators committed to the system	Actual generator outputs		Simulation model's generator outputs	
	Active power (MW)	Reactive power (MVAR)	Active power (MW)	Reactive power (MVAR)
Kukule-1	26	8.61	28.16	23.55
Kukule-2	26	8.61	28.16	23.55
New Laxapana-1	18.34	−0.421	19.6	16.86
New Laxapana-2	30	−0.278	19.6	16.86
Victoria-1	35	25.67	38.51	43.14
Victoria-2	50	26.65	56.22	41.27
Victoria-3	35	25.67	38.5	43.14
Ukuwela	20	10	20.71	12.44
Randenigala	48	20	44.49	32.64
Wimalasurendra	25	0	23.48	10.64
Laxapana-1	28.5	−3	24.92	15.39
Laxapana-2	22.5	−2	15.63	16.65
Samanalawewa-1	40	0.17	37.43	9
Samanalawewa-2	40	0.17	37.43	9
Bowatenna	11	9.23	8.78	5.36
Rantembe-1	15	1.74	24.1	14.6
Embilipitiya	44.3	30	38.48	24.38
Total	1607.4	705.52	1589.23	904.48

3.4.2 Generator Tripping/Sudden Generation Deficit Situation

It is necessary to see whether the simulation model responds to some emergency situation same as the power system faces that. If the actual results are very much closer to the simulation model's results, then it can be suggested that the model developed could be used to see the power system behavior under different scenarios. Norochcholai Coal Power plant tripping, which occurred at a time closer to the considered system loading (day peak time on 13th May, 2013) that lead for a total black out was simulated and observed the system performance.

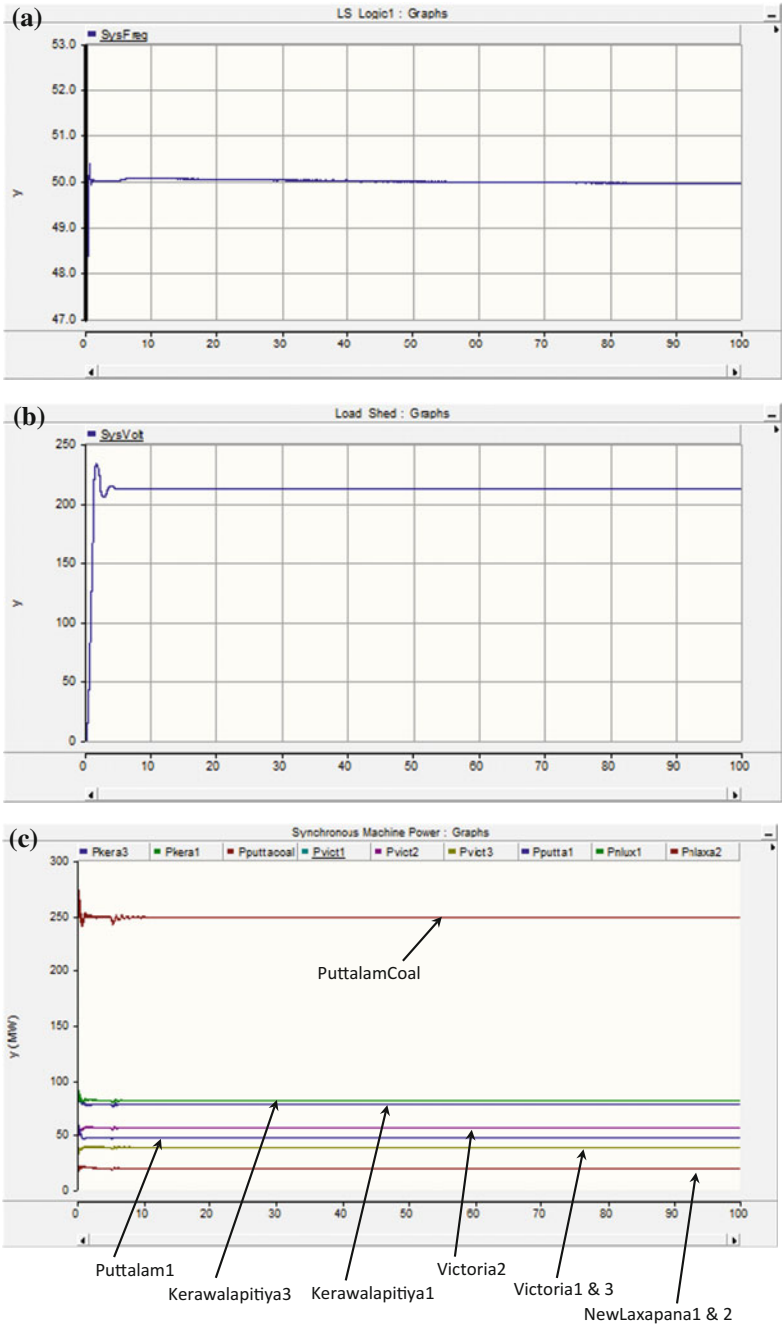


Fig. 3.51 Power system performance during steady state; **a** power system frequency, **b** power system (transmission) voltage, **c** some of the generator power outputs

Table 3.22 Removed generator outputs and loads from the PSCAD simulation model designed

Generators removed	Power delivered at the instant of the removal from the system		Loads removed	Power drawn at the instant of the removal from the system	
Victoria 1	37.35 MW	34.37 MVA	Biyagama	70 MW	30 MVA
Victoria 2	54.71 MW	32.64 MVA	Kotugoda	71.2 MW	25.4 MVA
Kerawalapitiya1	80.35 MW	2.6 MVA	Pannala	33.8 MW	21.8 MVA
KHD 1	16.66 MW	12.49 MVA	Aniyakanda	30 MW	18 MVA
KHD 2	16.66 MW	12.49 MVA			
TOTAL	205.73 MW	94.59 MVA		205 MW	95.2 MVA

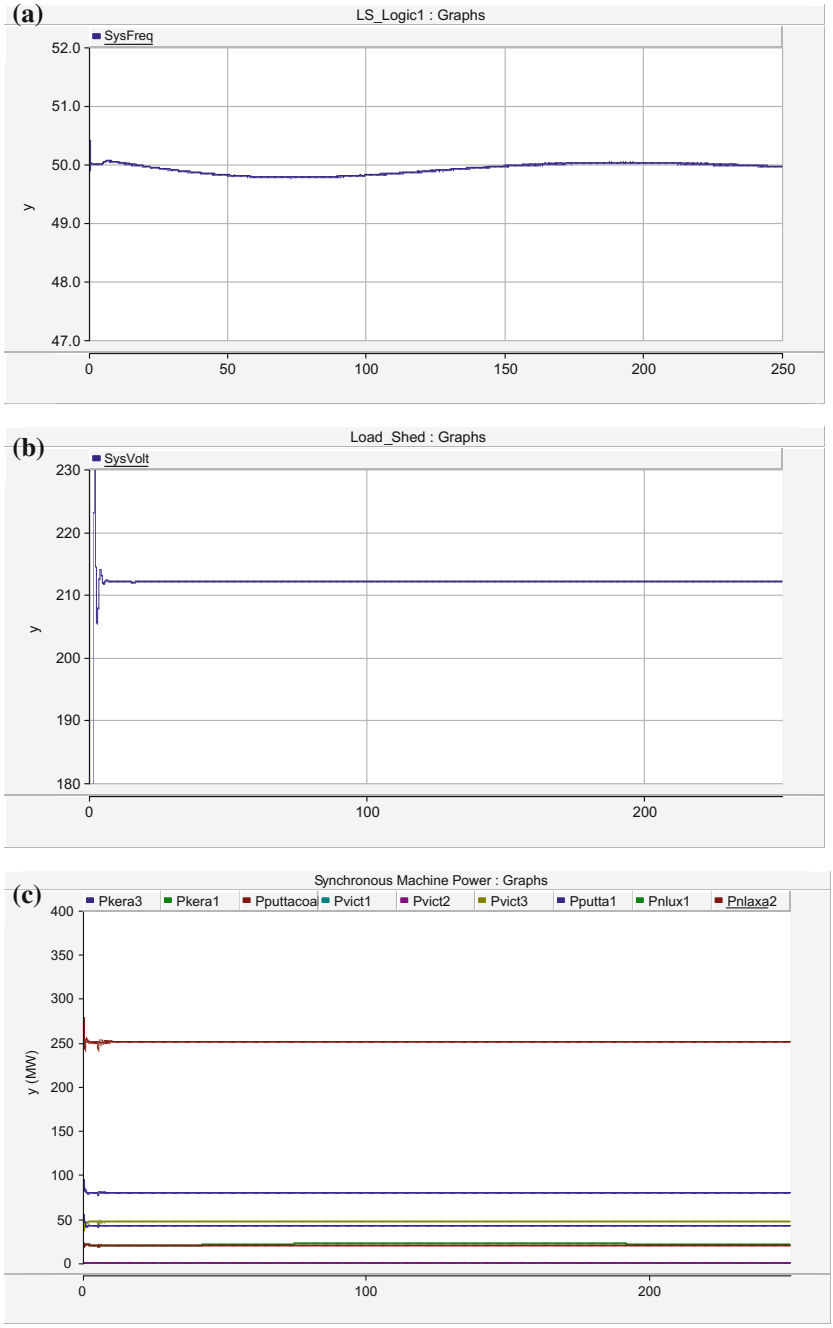


Fig. 3.52 After removal of some selected generator sets and loads (given in Table 3.22) from the simulated load flow, **a** system frequency; **b** system voltage; **c** power outputs of some generators

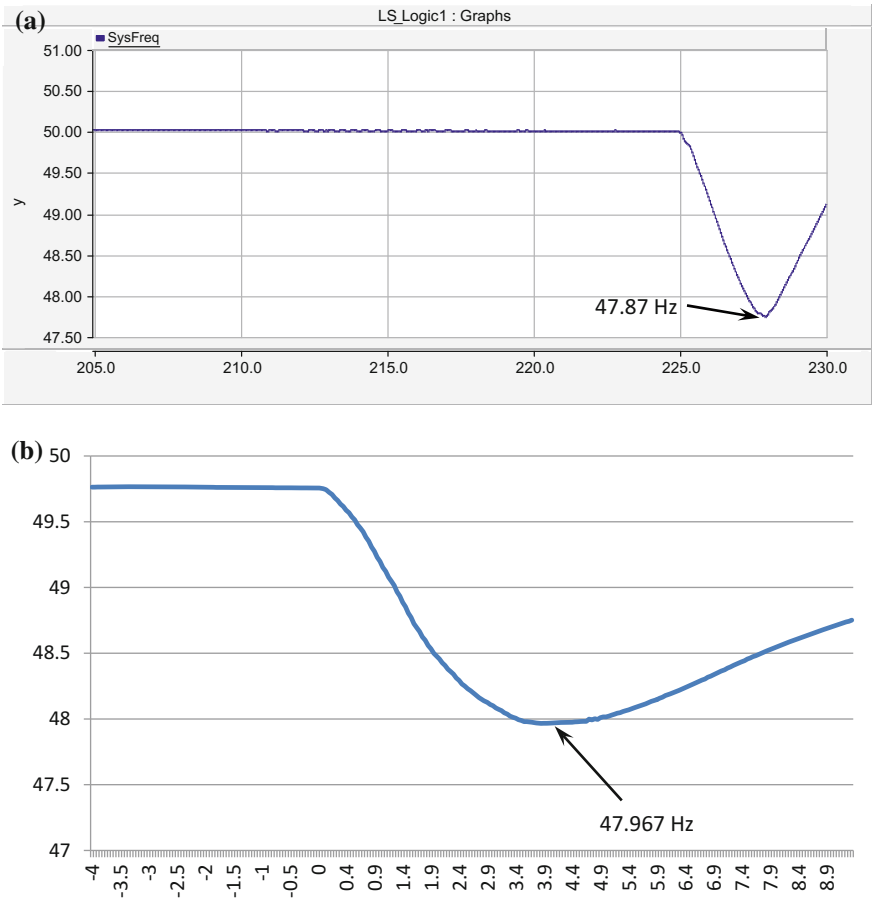


Fig. 3.53 Frequency profiles during a tripping of coal power plant; **a** with the simulation model (with some load and generation removal), **b** using actual system frequency variation data

The received data corresponds to an actual power system performance where the coal plant tripping occurs at a load flow of 1434 MW. Even though the coal power plant generates 300 MW, for its auxiliary services it consumes 25 MW. Therefore the balance 275 MW would be consumed by the system. This amounts 19.18% of the total demand. The already simulated model has a demand of 1637 MW. Therefor it has a difference of 203 MW (≃12.4% demand reduction) from the already modeled simulation program. If we try to simulate the same phenomena (i.e. coal power plant tripping) with this model the results we receive may be very much different to actual values. Therefor a simulation model with a load flow which has a demand of approximately 1437 MW is considered to compare the actual and simulation model results. To get the 1437 MW load flow, several power plants and loads were removed from the original simulation. The removed generators and

loads with their capacities are given in Table 3.22. Further, the coal power plant generates 250 MW with this load flow. It amounts 17.4% from the total demand.

With the above amendment, frequency, voltage and other selected generator outputs are shown in Fig. 3.52, under steady state condition.

Figure 3.53a shows the frequency profile when implement the load flow on 13th March, 2013 during peak load demand, with the amendments given in Table 3.22, while (b) shows the actual frequency profile during the generator tripping causing a generation deficit of 275 MW for the system. According to this comparison the simulation model reaches a minimum frequency of 47.87 Hz (a) where the actual system reaches a minimum frequency of 47.967 Hz (b).

So with the results shown in Fig. 3.53, it can be considered that the simulation model can give results much closer to the actual power system performances.